



Contents lists available at ScienceDirect

Journal of Pediatric Surgery

journal homepage: www.sciencedirect.com/journal/journal-of-pediatric-surgery

Review Article

Best Practices for Vessel Management in Pediatric Extracorporeal Membrane Oxygenation Cannulation, Decannulation, and Follow-up: A Narrative Review

Katherine M. McDermott ^a, Mohammed Moursi ^b, Sandra Tomita ^c, David H. Rothstein ^{d,*}^a Department of Surgery, Johns Hopkins School of Medicine, Baltimore, MD, USA^b Department of Vascular Surgery, University of Arkansas for Medical Sciences, Little Rock, AR, USA^c Division of Pediatric Surgery, Hassenfeld Children's Hospital at NYU Langone, NYU Langone Grossman School of Medicine, New York, NY, USA^d Division of Pediatric Surgery, Seattle Children's Hospital, University of Washington School of Medicine, Seattle, WA, USA

ARTICLE INFO

Article history:

Received 31 July 2024

Received in revised form

5 September 2024

Accepted 17 September 2024

Keywords:

Extracorporeal membrane oxygenation (ECMO)

ECLS

Extracorporeal support

Iatrogenic vascular injury

Vascular access

ABSTRACT

Extracorporeal membrane oxygenation (ECMO) to support neonates and children with cardiopulmonary failure was first described in the 1970s, since which time its use has expanded to an increasingly complex and heterogeneous pediatric population. Despite preserved survival outcomes, complications of ECMO use, including iatrogenic vascular injury, are common. Here, we provide a brief overview of the epidemiology of pediatric ECMO and associated vascular complications; describe common peripheral cannulation equipment and techniques, trends in cannulation and decannulation strategies, and respective incidence of vascular complications; and review existing evidence for best practices in cannula site selection, cannulation technique, decannulation strategies, and management of vascular complications, with the goal of providing a comprehensive review for interventionalists involved in the care of pediatric ECMO patients. Areas of wide practice variation in vessel management—application of vessel-sparing cervical venoarterial cannulation, the use of distal perfusion catheters in femoral arterial cannulation, and best practices for percutaneous single-lumen venovenous cannulation, as examples—areas of focus for future research, and the potential role of vascular surgeons and other subspecialty proceduralists in the care of pediatric ECMO patients are highlighted.

Level of Evidence: V.

© 2024 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Extracorporeal membrane oxygenation (ECMO) is a life-preserving therapy for patients with potentially reversible cardiac and/or respiratory failure. Pediatric patients supported by ECMO comprise a population of neonates to young adults with a wide range of congenital and acquired etiologies of cardiopulmonary failure [1,2].

Abbreviations: ECMO, extracorporeal membrane oxygenation; ELSO, The Extracorporeal Life Support Organization; VA, venoarterial; VV, venovenous; ECLS, extracorporeal cardiopulmonary resuscitation; VVV, venovenovenous; VVA, venovenovenous; VVDL-V, venovenous dual-lumen with additional cephalic venous drainage; IVC, inferior vena cava; NIRS, near infrared spectroscopy; DPC, distal perfusion catheter.

* Corresponding author. Department of Pediatric General and Thoracic Surgery, University of Washington School of Medicine, Seattle Children's Hospital, 4800 Sand Point Way NE, OA.9.220, Seattle, WA 98105, USA. Tel.: +1 206 987 5551; fax: +1 206 985 3378.

E-mail address: david.rothstein@seattlechildrens.org (D.H. Rothstein).

Data on risk factors for short- and long-term ECMO complications in children are limited by small sample sizes, heterogeneous patient populations, differences in procedural technique and medical management, and paucity of long-term follow up data. In light of limited evidence, no consensus guidelines exist to support best practices in cannulation or decannulation strategies [3,4], and there is wide practice variation in cannulation and decannulation technique among pediatric ECMO practitioners [4–6]. This review will provide a brief overview of the epidemiology of pediatric ECMO and associated vascular complications; summarize ECMO equipment, ECMO modalities and their respective indications; describe common peripheral cannulation techniques, including trends in cannulation strategies and incidence of associated vascular complications; summarize existing evidence for best practices in cannula site selection, cannulation technique, and decannulation strategies; and highlight knowledge gaps and areas for future research.

<https://doi.org/10.1016/j.jpedsurg.2024.161961>

0022-3468/© 2024 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

1. Epidemiology of vessel management and vascular complications in pediatric ECMO

The earliest reported cases of ECMO use in children were in the 1970s, first for post-cardiotomy failure in an infant unable to wean from cardiopulmonary bypass in 1972 and, later, for a neonate with respiratory failure secondary to meconium aspiration syndrome [7]. ECMO use in infants and children had increased steadily over the 40 years following these first cases, and many current practices in both pediatric and adult ECMO are extrapolated from early neonatal data [8]. Approximately 800–1000 cases of neonatal (age 0–28 days) and pediatric (age 29 days to 18 years) ECMO are now performed each year in the United States [9,10]. Median annual neonatal and pediatric ECMO cannulation volumes range from 5 to 10 cases per center per year, and high volume North American centers typically perform 50 cases per year or more [11]. Most practicing pediatric surgeons perform fewer than 5 cannulations per year [4–6]. Despite relatively low physician- and center-volumes and increasingly high patient acuity, global survival to ECMO decannulation among neonates and children remains consistently above 70%: 70% and 87% in neonates for cardiac and pulmonary indications, respectively, and 72% in pediatric patients for either indication [11].

In the context of increasing institutional and national experience, improved ECMO survival, and expanded ECMO use in older pediatric populations, efforts to characterize and minimize the short- and long-term complications of ECMO cannulation gained attention in the early 1990s and remain a topic of interest for surgeons and intensivists. The Extracorporeal Life Support Organization (ELSO), an international clinical and research collaborative, was founded in 1989 to collect and disseminate data on ECMO practices and outcomes. As of 2022, ELSO membership includes 583 centers, the majority of which are in North America, and it has amassed a registry of more than 87,000 neonatal and pediatric ECMO runs [11].

ELSO facilitates large multi-institutional studies, which are often used to develop best-practice recommendations, but the literature on vessel management and vascular complications in pediatric ECMO remains dominated by single-institution case series and retrospective analyses. Detailed information on cervical vessel management are not included in ELSO data, and although lower extremity vascular complications have been explicitly collected by the ELSO registry since 2013, the incidence of these complications in ELSO is much lower than in published institutional data, likely reflecting underreporting or incomplete capture [12,13].

In-hospital vascular complications, including cannula site bleeding, arterial injury requiring intervention, clinically significant deep venous thrombus, or embolus or acute ischemia (e.g., stroke or acute limb ischemia) occur in 7%–54% of neonatal and pediatric ECMO patients and are associated with increased risk of mortality [14–16]. The reported incidence of vascular complications varies by cannulation site and ECMO modality, with lowest rates in internal jugular venovenous (VV) cannulation, and highest in peripheral venoarterial (VA) cannulation [14].

Mid- and long-term vascular complications include neurologic and/or cerebrovascular consequences of carotid artery cannulation in neonates and children [17,18] and late lower extremity peripheral vascular complications following femoral cannulation. Several single-center series with mid- and/or long-term vessel imaging follow-up have demonstrated prevalence of persistent carotid arterial narrowing or wall defects after vessel sparing or reconstruction to be as low as 0% and as high as 50% [19,20]. One series examining adolescents with a history of cervical vessel ligation in infancy found contralateral (left) carotid intimal thickening in ECMO survivors compared to age- and sex-matched peers [21]. The

implications of these ipsilateral or contralateral abnormalities for risk of clinically significant cerebrovascular disease into adulthood remain unclear. Late lower extremity peripheral vascular complications, including cannula site arterial stenosis or subtle lower extremity growth and developmental abnormalities following femoral artery cannulation, have been described in small retrospective series of children with a history of diagnostic femoral artery cannulation but are not well-studied after pediatric ECMO [22].

The focus of the current review is the practice of peripheral ECMO, including cervical, femoral, and in rarer cases brachial, axillary, or subclavian vessel cannulation. In current practice, central ECMO cannulation—typically involving placement of a venous catheter in the right atrium, an arterial catheter in the ascending aorta, and, if necessary, an additional venous drainage catheter in the left atrial appendage to reduce left-sided heart pressures [23]—is primarily reserved for post-cardiotomy patients, or for emergent cardiopulmonary support in children with congenital cardiac anomalies as a bridge to more definitive interventions. Small retrospective case series suggest that the higher flows achieved with central cannulation may improve outcomes in high-flow states like sepsis [24,25]; however, there is insufficient evidence to routinely support this approach, and practice has not shifted to use central cannulation for refractory septic shock in non-cardiac patients. While many of the considerations below will apply to central ECMO, the remainder of this review is meant to describe equipment, techniques, and complications specifically relevant to peripheral ECMO cannulation.

2. Brief overview of ECMO circuit and cannulas

2.1. Circuit

A standard ECMO circuit comprises three primary components: a mechanical pump, a membrane oxygenator for gas exchange, and a heat exchanger to maintain euthermia [26] (Fig. 1). The same core circuit is used for both VA and VV ECMO. Additional ports for a second return (venous drainage) cannula or outflow cannula (e.g., if separate arterial outflow is required for a distal perfusion catheter), or a bridge between return and outflow cannulas can be added as needed. A detailed discussion of the components and evolution of the ECMO circuit is beyond the scope of this paper, but it is worth noting that technological advances in component parts and surface coatings in the 1990s and 2000s improved heat and gas exchange efficiency, reduced circuit thrombogenicity, and have contributed to better ECMO survival outcomes [27].

2.2. Peripheral ECMO configurations and cannula types

Indications for ECMO use in the pediatric population are broad but are generally divided into primary respiratory failure, primary cardiac failure, “other” or mixed cardiopulmonary failure, and extracorporeal cardiopulmonary resuscitation (ECPR) in the setting of refractory cardiac arrest (Table 1) [11,26,28]. VV ECMO, in which blood is both drawn from and returned to the patient’s central venous circulation, most commonly the vena cava and right atrium, can support patients with severe respiratory failure and preserved cardiac function. Peripheral VA ECMO similarly diverts blood from central venous circulation, but then returns blood to aortic circulation. The physiologic benefits of VV over VA ECMO use in appropriate candidates include the return of more highly oxygenated blood to pulmonary and myocardial vasculature, maintenance of pulsatile blood flow, lower risk of arterial emboli (which are filtered by the pulmonary circulation), and avoiding complications associated with peripheral arterial access [28,29]. In neonates and small children, jugular vessel size may not accommodate larger

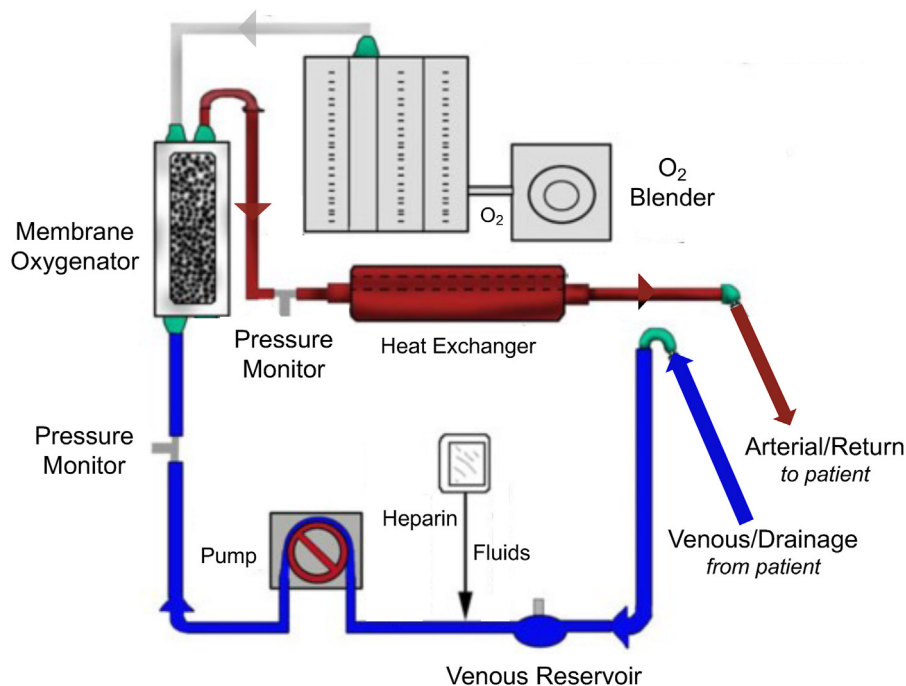


Fig. 1. Standard extracorporeal membrane oxygenation circuit components. Adapted from Kilic, A. Extracorporeal Membrane Oxygenation, 2018 [85].

dual-lumen cannulas, and two-site access for VV ECMO is limited by small femoral vessel size and increased risk of recirculation, in which return cannula flow is preferentially directed into the drainage cannula and returned to the circuit rather than to the patient's native circulation.

Hybrid or triple cannulation strategies are used infrequently in children. In patients supported by VV ECMO who develop cardiac failure, an arterial cannula can be added to the existing ECMO circuit without the need to replace existing single or dual-lumen venous cannulas. In cases of insufficient venous drainage after initial cannulation, an additional femoral or cephalic venous cannula can be used to supplement venous inflow (VVV or VVA); some centers advocate for routine use of an additional cephalic venous drainage cannula in neonates undergoing VV ECMO through a dual-lumen cannula (VVDL-V [30]).

The majority of neonatal ECMO is performed for respiratory indications, though its use in ECPR and "other" indications has increased over the past decade [9,11]. VA ECMO comprises greater than 90% of ECMO use in neonates and remains the most common modality for primary respiratory failure in this population. Among older infants and children, the proportion of ECMO performed for primary cardiac indications or as part of ECPR is also increasing, and these cases now represent the majority of pediatric ECMO runs [11]. Unlike in neonates, there is a slight VV predominance for pure respiratory failure in pediatric patients 29 days and older (60%–70% of ECMO runs, with higher proportions in adolescent patients), but VA is still frequently used for primary respiratory indications in smaller children.

2.3. Cannula features and sizes

Cannula types can be broadly divided into single lumen and dual-lumen configurations. Single lumen cannula designs include those specific to venous drainage (circuit inflow) and arterial return (circuit outflow), respectively, and those that can be used for either arterial or venous flow (Fig. 2). In general, venous drainage cannulas require more side holes along the distal catheter to allow for

distribution of negative pressure. Lighthouse or semi-lighthouse tip cannulas, which have a central hole at the catheter tip with additional rows of side holes along the distal catheter, can be used for either venous drainage or arterial return; blunt tip cannulas, which have a single lumen and central hole at the distal catheter tip, are used exclusively for arterial return (Fig. 3). Common pediatric single lumen cannula types are available in sizes from 8Fr to 29Fr (Table 2) [31,38].

Dual-lumen cannulas for single-site VV ECMO exist in bicaval (e.g., Avalon Elite [Avalon Laboratories, Rancho Dominguez, CA] and right atrial (e.g., Crescent [Medtronic, Minneapolis, MN]) variations, each with interrupted rows of cannula side holes for venous drainage—typically both proximal to and distal to a single return port designed to direct return flow toward the tricuspid valve (Fig. 2). Specific concerns about high complication rates, specifically migration, clinically significant pericardial effusions, and right atrial perforation, following Avalon Elite bicaval cannula use in children surfaced shortly after the introduction of pediatric-sized wire-reinforced Avalon cannulas in 2009 [32,33]. Subsequent single-center studies adjusting for patient characteristics and a large retrospective review of ELSO registry data have not demonstrated higher risk of pericardial effusion or severe cardiac complications requiring intervention associated with bicaval/wire-reinforced dual-lumen cannulas compared to atrial dual-lumen cannulas or multisite VV ECMO, and the risks of severe tamponade with bicaval cannulation appear to be mitigated by use of both echocardiographic and fluoroscopic image guidance during placement [34–37].

Re-circulation of newly oxygenated blood directly into venous drainage ports is a particular challenge associated with the use of dual-lumen cannulas in small patients, especially in bicaval designs, and size of the patient's thoracic cavity is an important consideration in VV dual-lumen cannula selection. The smallest currently available dual-lumen cannulas are 12Fr (right atrial) and 13Fr (bicaval), and larger adult dual-lumen cannulas are available in sizes up to 32Fr [31]. In neonates with respiratory failure whose vessels are unable to accommodate the smallest dual-lumen

Table 1A
Common Indications for Peripheral ECMO use in Neonatal and Pediatric Populations [26,28,83].

Indication	Common Etiologies	Mode
Respiratory		
Neonatal	Meconium aspiration syndrome	VA or VV*
	Congenital diaphragmatic hernia	VA or VV*
	Congenital pulmonary or airway malformation	VA or VV*
Neonatal or pediatric Airway	Management of difficult airway	VA or VV*
	Upper airway obstruction	VA or VV*
	Mediastinal mass	VA or VV*
Lung/Respiratory	ARDS	VA or VV*
	Pneumonia	VA or VV*
	Bronchopleural fistula	VA or VV*
	Severe pulmonary contusion	VA or VV*
	Alveolar hemorrhage	VA or VV*
	End-stage lung disease awaiting lung transplant	VV
Cardiac		
Neonatal or pediatric	Structural congenital heart disease	VA
	Cardiomyopathy	VA
	Myocarditis	VA
	Refractory arrhythmia	VA
	Heart failure	VA
	Myocardial infarction	VA
	Pulmonary hypertensive crisis	VA
	Massive pulmonary embolus	VA
	Cardiac contusion	VA
	Failure to wean from CPB	VA
Other/Mixed		
Neonatal or pediatric	Sepsis	VA or VVA
	Toxic Ingestion	VA
	Hypothermia	VA
ECPR †		
Neonatal or Pediatric	Sustained in-hospital cardiac arrest (or witnessed out of hospital arrest with short transport time and high-quality compressions), refractory to conventional CPR	VA

ECMO, extracorporeal membrane oxygenation; VV, venovenous; VA, venoarterial; ARDS, acute respiratory distress syndrome; VAD, ventricular assist device; CPB, cardiopulmonary bypass; ECPR, extracorporeal cardiopulmonary resuscitation; CPR, cardiopulmonary resuscitation.

* VV is feasible in many patients, but access for VV may be limited by patient size, required or available cannula sizes, and recirculation; VVV with additional cephalic drainage is one option for addressing limited inflows in this population, but VA is often used for primary respiratory failure in neonates and small children.

† Should be limited to experienced ECMO centers.

catheters, or whose thoracic cavities are too small to allow for adequate positioning of dual-lumen VV cannulas without significant recirculation, VA cannulation is often used even in the absence of significant cardiac dysfunction.

Table 1B
Contraindications to ECMO use in Neonatal and Pediatric Populations [26,28,83].

Absolute	Relative
Lethal chromosomal or syndromic abnormalities (e.g., trisomy 13 or 18)	Weight <2 kg
Extreme prematurity*	Gestational age <34 weeks*
Irreversible neurologic injury	Prolonged duration of mechanical ventilation
Irreversible organ injury without the possibility of transplant	Intracranial hemorrhage or recent (within 7d) neurosurgical procedure
Incurable malignancy	Preexisting chronic illness with poor prognosis
	Coagulopathy

* Recommended thresholds for gestational age as an absolute contraindication vary, and use in premature infants <34 weeks has increased in the last decade [84].

Venous drainage is the rate-limiting factor for ECMO circuit pump flow, and the goal is to use the largest diameter, shortest venous cannula that is anatomically feasible. An oversized cannula risks vessel damage, especially to veins, and an undersized cannula may provide inadequate flows. Nomograms to estimate cardiac output for patient weight are used to determine required circuit flows and, in turn, minimum acceptable cannula sizes for a given cannula length and modality [38]. Flow targets are near 100 mL/kg/min in most patients; reference flows range from 80 mL/kg/min to 200 mL/kg/min based on patient size and cannula position. Flow needs may be higher than nomogram estimates in distributive shock or lower in pure respiratory failure with preserved cardiac function.

3. General anatomic and technical considerations for cannulation

The indication for ECMO, modality, patient age, weight and vessel size, previous line history and anticipated need for future lines or repeat ECMO are the primary considerations in choosing a peripheral cannulation site and strategy.

3.1. Selecting a peripheral cannulation site

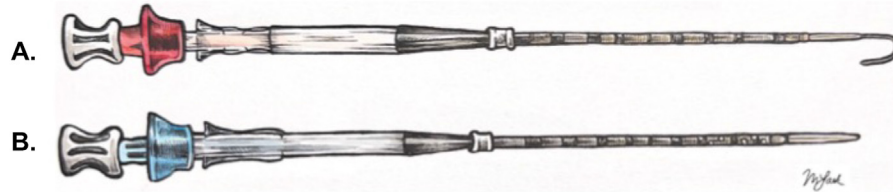
Right cervical vessels and bilateral femoral vessels are the most accessed sites for peripheral ECMO cannulation in children (Fig. 4, Fig. 5) [23,39]. Alternative peripheral sites (e.g., axillary or subclavian vessels) have been proposed as an option to preserve ante-grade bilateral carotid circulation during and after ECMO but are less frequently used in pediatric populations than in adults [40]. Extraperitoneal iliac vessel or inferior vena cava (IVC) cannulation in infants with complicated access needs have been described but are rarely performed [3,41].

Cannulation in neonates and small children is typically performed via an open cervical approach [3,11]. Though VV ECMO, either by an open or ultrasound-guided percutaneous approach, is feasible in neonates, cervical VA still comprises an estimated 85%–90% of ECMO in this group [3,42]. Femoral cannulation is the dominant site of cannulation in adults undergoing peripheral ECMO, but small vessel size and longer intracorporeal catheter length limit achievable flows in neonates and increase the risk of ipsilateral limb ischemia in children. The decision to use femoral vessels for pediatric cannulation in school-aged children and adolescents remains heavily influenced by practitioner and center preference and experience [6]. Various age and weight cutoffs for considering femoral cannulation have been proposed, ranging from >2 years to >7 years and from >10 kg to >30 kg, but survey data suggest femoral cannulation is not routinely performed by practicing U.S. general pediatric surgeons in children <13 years or <35 kg [5,41,43,44].

Duplex ultrasound is a useful tool to characterize vessel size, examine for stenosis or thrombus, and evaluate for local hematoma at sites with existing central lines, any of which might affect the optimal approach to cannulation. This is especially relevant in patients with unknown line history, with previous or current indwelling central venous catheters at potential cannulation sites, or in whom both cervical and femoral cannulation are theoretically feasible. Echocardiograms are often available in patients under consideration for ECMO, but in the absence of available echocardiography imaging, evaluating for anatomic variants like persistent left-sided superior vena cava or other anomalous venous anatomy may be worthwhile if the right internal jugular vein is smaller than expected [41].

It is worth noting that existing literature is mixed on the benefits of avoiding cervical cannulation in favor of femoral, axillary/

Single-Lumen Cannulas



Double-Lumen Cannulas

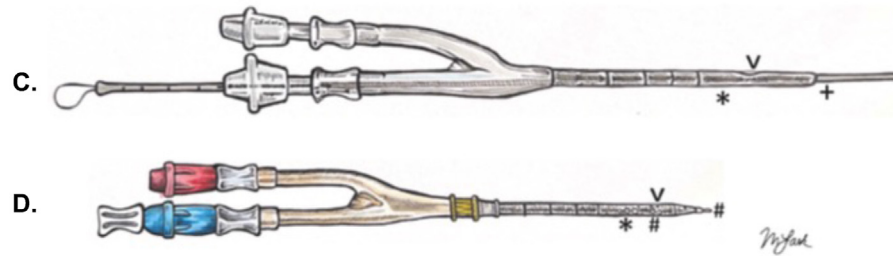


Fig. 2. Common Extracorporeal Membrane Oxygenation Cannula Types. A. Arterial (return) cannula B. Venous (drainage) cannula, for venoarterial or dual-site venovenous ECMO C. Avalon Elite™ dual-lumen bicaval cannula D. Crescent™ RA cannula; SVC drainage (*), Right Atrial drainage (#), IVC drainage (+); Right Atrial reinfusion (v) directed toward the tricuspid valve. Adapted from Jensen et al. 2022 [48].

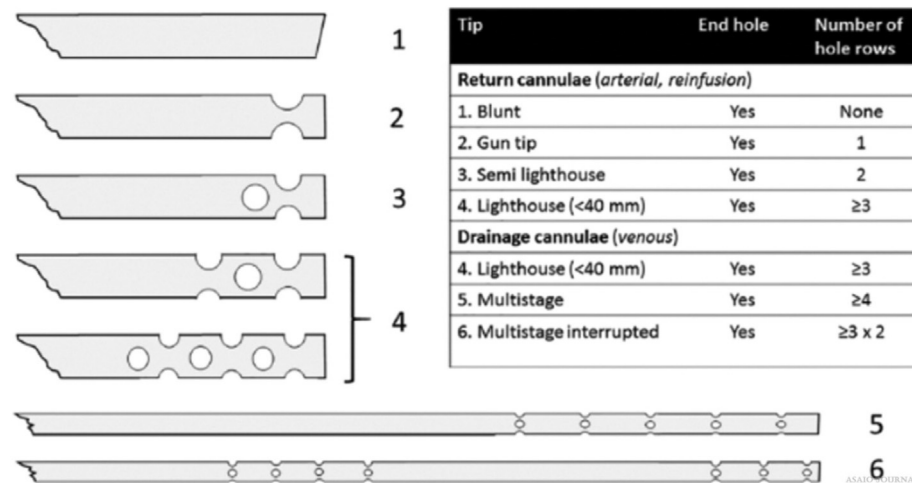


Fig. 3. Tip Features of Commonly Used Return and Drainage Cannulas. Gajkowski et al. 2022 [23].

Table 2

Pediatric venoarterial and venovenous ECMO cannula sizing by weight [31,38].

Patient Weight	Arterial Cannula Size		Venous Cannula Size		
	Short (cervical/axillary)	Long (femoral)	Single lumen Short (cervical)	Long (femoral)	Dual-lumen (cervical)
<3 kg	8-10 Fr	—	12-14 Fr	—	12-13 Fr
3–10 kg	10-12 Fr	—	14-16 Fr	—	12-16 Fr
10–20 kg	12-14 Fr	12-15 Fr	14-18 Fr	15-19 Fr	19 Fr
20–30 kg	14-16 Fr	12-15 Fr	16-20 Fr	19-21 Fr	19-20 Fr
30–40 kg	16-18 Fr	17-19 Fr	20-22 Fr	19-23 Fr	19-23 Fr
>40 kg	18-20 Fr	19-23 Fr	22-24 Fr	23-27 Fr	27-31 Fr

subclavian, or central cannulation in children [3]. Di Gennaro et al., in a database analysis of the ELSO registry, found a 4-fold increase in stroke events in patients aged 6–40 when comparing carotid to femoral cannulation [46], in contradistinction to a similar ELSO study by Johnson et al., which found that in the 0–18 year old age

group, no significant differences existed in neurologic outcomes between different cannulation sites [3]. Some contemporary cohorts strongly suggest VV ECMO is associated with lower rates of acute intracranial hemorrhage or infarct [30,45]. Neonates have excellent cerebrovascular collateral circulation, but the biological or

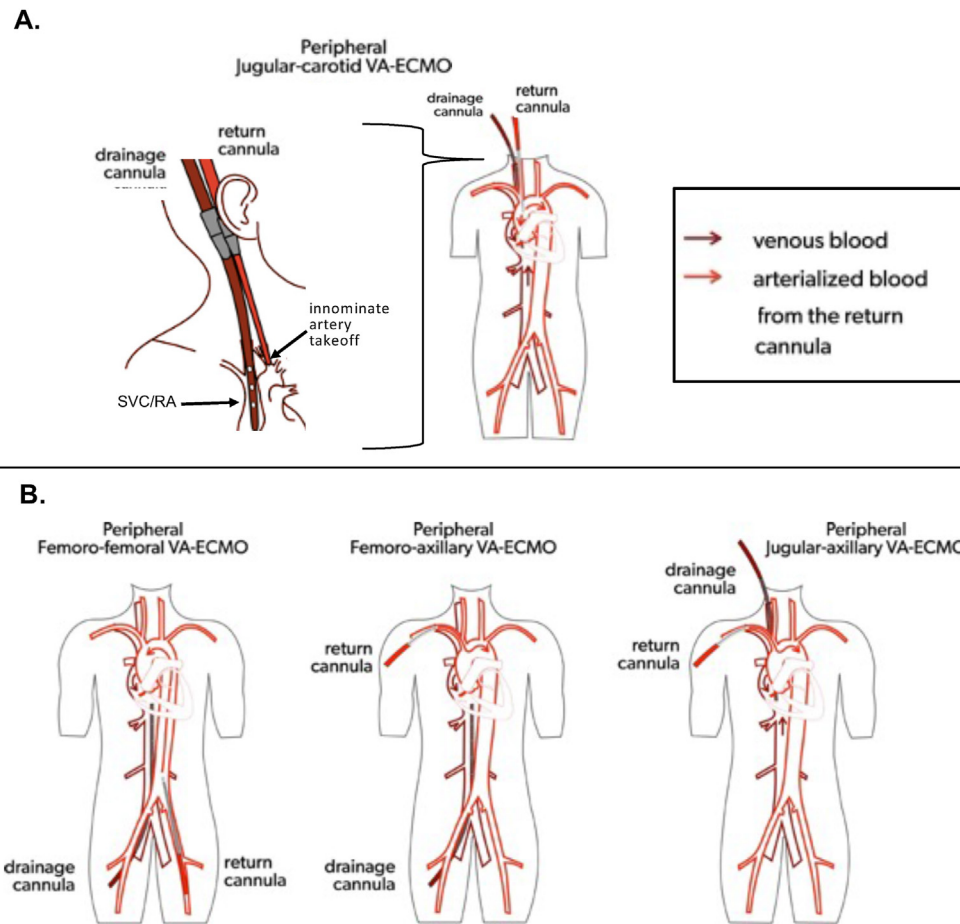


Fig. 4. Common peripheral venoarterial (VA) cannula configurations in neonates and children. (A) VA ECMO cannula configuration via cervical access in neonates and small children: a cannula placed in the right internal jugular vein drains from the right atrium, and a return cannula placed in the right carotid artery returns oxygenated blood to the innominate artery just distal to its takeoff from the aortic arch. (B) Alternative configurations involving femoral access used in larger children and adolescents. Gajkowski et al. 2022 [23].

physiological age at which cervical cannulation increases risk of acute cerebrovascular complications relative to other cannulation sites is not well-understood. Some authors have suggested 5 or 6 years of age and >20 kg as cutoffs for preferentially pursuing femoral cannulation over cervical cannulation, but even recent studies using overlapping ELSO samples have found conflicting results about the relative stroke risks by cannulation site [3,46,47].

3.2. Open, percutaneous, and hybrid cannulation approaches

Open, percutaneous, and hybrid or cutdown strategies are routinely used for both cervical and femoral ECMO cannulation. No studies in children have prospectively compared these approaches for any site, but safety and success rates with percutaneous VV or VA access appear similar to open cannulation in observational studies [42,43].

Open technique has been associated with higher rates of mechanical cannula/circuit complications, cannula site bleeding, and vascular repair compared to percutaneous cannulation in retrospective cohort studies, but it is difficult to account for confounding by indication: planned vessel preservation/reconstruction or concerns about mechanical complications due to body habitus, existing hematoma from previous lines, or other anatomic factors that might predict difficult percutaneous access are themselves reasons to pursue open cannulation [42]. Obesity and coagulopathy may

increase the difficulty and bleeding risk associated with open cannulation. It is not surprising that retrospective reviews of ELSO data confirm that percutaneous access is more frequently used in larger, older children [42]. Although hybrid/cutdown strategies are frequently described in technique papers or expert recommendations, they are not reliably captured by existing registries, which limits even retrospective comparisons between hybrid and other methods [13,43,48,49].

4. Cervical ECMO cannulation, decannulation, and vessel management

4.1. Cannulation

Cervical cannulation in neonates is performed via an open approach in most cases (Fig. 6). This technique involves a transverse incision 1 to 2 fingerbreadths above the clavicle over the anterior border of the sternocleidomastoid, division of the subcutaneous tissue and platysma with electrocautery to minimize cannula site bleeding, and blunt separation of the heads or retraction of the sternocleidomastoid to expose the carotid sheath. The carotid sheath is opened sharply. The right internal jugular vein is then gently dissected circumferentially, taking care to avoid jugular vasospasm, and silk ties are loosely placed proximally and distally to the planned cannulation site. The right common carotid artery is

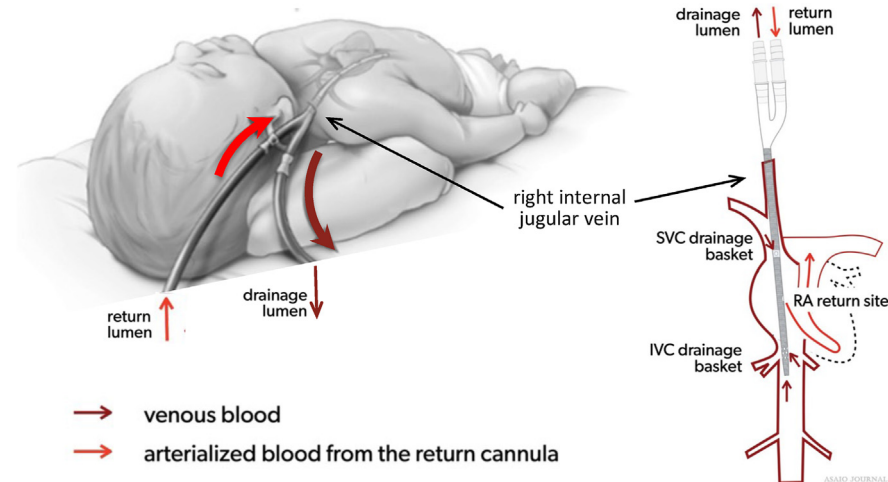
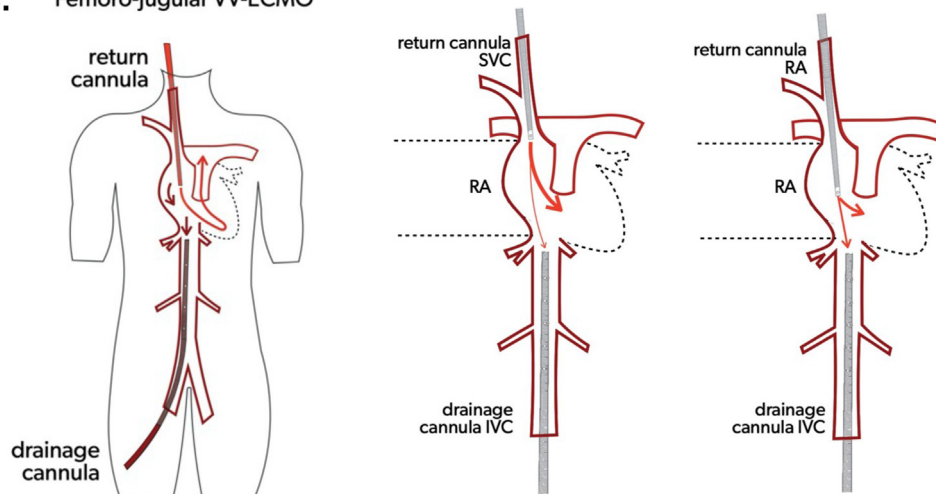
A. Single-Site Jugular VV-ECMO**B.** Femoro-Jugular VV-ECMO

Fig. 5. Common venovenous (VV) cannula configurations in neonates and children. (A) Single-site VV ECMO using a dual-lumen catheter placed in the right internal jugular vein, draining from the superior vena cava, right atrium, and IVC, reinfusing into the right atrium with the reinfusion port oriented toward the tricuspid valve. (B) Two-site VV ECMO using two single-lumen cannula, a drainage cannula placed in either femoral vein and draining from the distal IVC/RA, and a reinfusion cannula placed in the right internal jugular vein, reinfusing into the RA with the reinfusion port oriented toward the tricuspid valve. Adapted from Gajkowski et al. 2022 [23].

then dissected over a similar course and proximal and distal silk ties are placed. Care is taken to avoid injury to/entrapment of the vagus nerve. Once sufficient exposure has been achieved, catheter selection based on vessel diameter(s) can be confirmed.

Intravenous heparin is administered (typically 100U/kg, though some experts recommend half-dosing for patients with existing coagulopathy). A waiting period of 3 min is standard to allow the full therapeutic effect—alternatively, monitoring for an ACT of 200 can be performed, but this is not routine at most centers. This period can be used to measure and mark catheter(s) at the ideal insertion length.

Classically, both the common carotid artery and internal jugular vein are ligated cephalad to the cannula insertion sites at the time of cannulation using the pre-placed silk ties, a step thought to prevent cerebral thromboembolism during ECMO support. After ligation, the vein and artery are sequentially cannulated via transverse venotomy/arteriotomy through the anterior vessel wall, then secured to each vessel with silk ties cushioned by a rubber bumper (Fig. 6).

A vessel-sparing approach using pre-placed purse string sutures is also described: this technique involves placing an oblong purse string suture in the anterior vessel wall using 5–0 or 6–0 Prolene,

performing a venotomy/arteriotomy in the center of the purse string, and inserting the cannula through the purse string site, then securing the cannula with a Rummel tourniquet (Fig. 7). Some centers have instituted this routinely for neonatal and pediatric cervical cannulation [19,20].

Small case series, first in 2 neonates in the early 1990s and more recently in 9 children and adolescents, report carotid artery cannulation via synthetic graft to the right common carotid artery (a “chimney graft”) [50,51]. In this technique, cervical vessels are exposed, and systemic anticoagulation is administered as described above. The internal jugular vein is cannulated first, after which the common carotid artery is clamped, and a synthetic graft is sewn to the artery in an end-to-side fashion (Fig. 8). The arterial cannula is then inserted through the graft, with the tip either at the graft-arterial anastomosis site or passed distally to the standard position at the junction of the innominate artery and the aorta, and the cannula is secured to the graft using circumferential ties. In both series, the grafts were suture-ligated near the anastomosis at the time of decannulation. Without clear benefit over other vessel-sparing techniques, and due to both the time-consuming

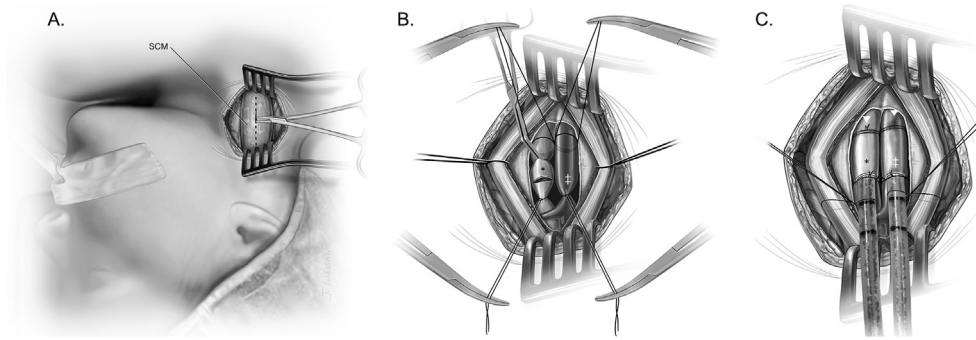


Fig. 6. Open Right Cervical Venoarterial ECMO Cannulation. Traditional open approach involving distal ligation of both the internal jugular vein and common carotid artery at the time of cannulation. (A) A transverse incision is made over the right neck to expose the sternocleidomastoid (SCM), which is then divided longitudinally over the carotid sheath. (B) The right common carotid artery (*) and internal jugular vein (‡) are encircled distally and proximally with heavy silk sutures, and following heparin administration the distal arterial and venous ties are ligated and tails left long for retraction and cannula securement. (C) First the vein and then artery are sequentially cannulated using transverse venotomy/arteriotomy; each cannula is secured distally using the tail of the silk tie ligation and proximally by tying down the silk sutures over the cannulated vessel (white arrowheads). Though not pictured here, proximal ties are usually secured over a vessel loop bumper.

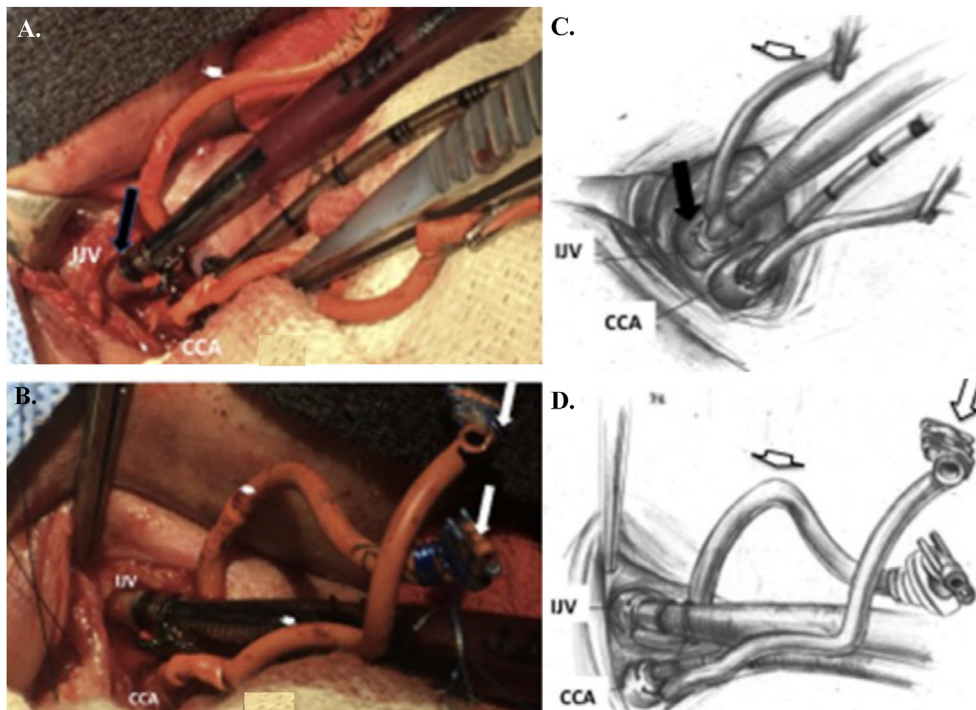


Fig. 7. Vessel-sparing Cervical Venoarterial ECMO Cannulation. Vessel-sparing cannulation technique. Intraoperative photos (A, B) and illustrations (C, D) demonstrating 5–0 Prolene purse-string sutures placed on vessels (Black Arrow), held in place using Rommel's tourniquets (white arrowhead). Tourniquets are secured with Hemoclips (white arrows). CCA: common carotid artery; IJV: internal jugular vein. Adapted from Kurkluoglu et al., 2016 [20].

anastomosis required at the time of cannulation and the risk of cerebral hyperperfusion, this technique has not been widely evaluated.

Percutaneous cervical cannulation is used with increasing frequency for VV ECMO in children [42,45]; successful percutaneous carotid cannulation for VA ECMO has been reported, but is uncommon [43,45]. Direct visualization of the cervical vessels using a cutdown approach as described above, with or without circumferential mobilization of the vessels, has not been directly compared to either open or purely percutaneous approaches, but single-center series have reported excellent success and safety of VV cannulation in neonates as small as 2 kg without cutdown [52]. Duplex evaluation of the cervical vessels and vena cava should be performed at the time of cannulation; modified Seldinger technique under ultrasound guidance is then used to access the internal

jugular and/or carotid vessels and dilate the subcutaneous tract. Given the generous size of ECMO cannulas relative to target vessels, the cannula(s) should be lubricated prior to placement—propofol and mineral oil are two widely available options for lubrication. Bolus anticoagulation before cannula placement remains indicated in the percutaneous approach, but reduced dosing (50U/kg of IV heparin) may be appropriate for percutaneous cannulation [43,52].

4.2. Decannulation, reconstruction, and catheter exchange

The standard approach to cervical ECMO decannulation in neonates and children is to ligate the carotid and jugular vessels caudal to the cannulation site at the time of cannula removal. Carotid preservation and/or reconstruction as an alternative to ligation has been investigated since the early 1990s as a way to

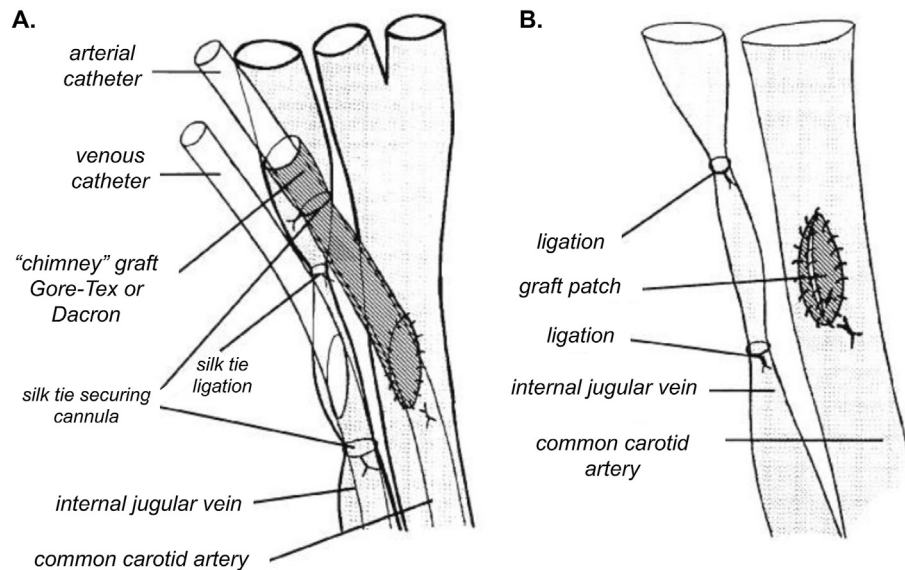


Fig. 8. Cervical Venoarterial ECMO Cannulation via End-to-Side Carotid Chimney Graft. (A) A prosthetic graft is sewn to the right common carotid artery in end-to-side fashion, and the arterial cannula is inserted through the graft and distally into standard position in the distal common carotid/proximal innominate artery. (B) At decannulation, the graft is closed to create an arterial patch, and the vein is ligated in standard fashion for a non-vessel-sparing technique. Dacar et al. 1992 [51].

facilitate repeat ipsilateral ECMO cannulation and prevent or mitigate acute stroke and long-term neurodevelopmental complications [53–55]. In the vessel-sparing technique described above, the pre-placed purse string suture is cinched and tied at the time of cannula removal. Alternative strategies include primary arteriotomy repair at the time of decannulation, segmental resection and primary anastomosis (for short-segment intimal injury, for example), and, less commonly, interposition graft or patch repair [56]. For the latter, autologous vein and bovine pericardium or carotid are all suitable conduits [48].

Internal jugular venous preservation and/or reconstruction are less common than arterial salvage. The vein is more susceptible to damage from ECMO cannula, especially as time on ECMO increases. Sacrificing the internal jugular vein is thought to be less clinically significant for cerebral perfusion, thus anticipated need for repeat ECMO cannulation is the primary consideration when pursuing venous preservation or reconstruction. Indications for continued alternative central access should also be considered in all patients who are planned for decannulation, regardless of cannulation strategy or anticipated risk of future ECMO. Exchanging the venous ECMO cannula for another long-term central venous line (e.g., a tunneled central line or hemodialysis catheter) is a safe option for patients who require long-term central access and does not appear to increase the risk of access site complications or central line associated infections compared to de-novo central line placement [48].

4.3. Practice patterns and cerebrovascular outcomes following cervical ECMO cannulation

There is wide practice variation in the management of cervical vessels during and following ECMO cannulation [4]. General pediatric surgeons most commonly ligate cervical vessels at decannulation, while congenital cardiovascular surgeons are more likely to reconstruct them [4,6]. As mentioned above, some general pediatric surgery practices have instituted vessel-sparing cannulation for all patients, though this is not widely considered standard of care [19,20].

Early and long-term vessel patency, subclinical cerebrovascular or intracranial imaging abnormalities, and short- and long-term

complications, including stroke, neurodevelopmental delays, and feasibility of repeat cervical cannulation, are the most frequently examined outcomes in studies examining the potential benefits of cervical vessel preservation.

Small initial case series of reconstruction reported 70–100% common carotid artery patency, no clinically significant acute embolic strokes, and normal short-term carotid and intracranial duplex exams [54,55,57]. More recent series examining reconstruction or vessel preservation report a broader range, from 44% to 100% common carotid artery patency [58]; this is compared to right internal carotid artery patency rates ranging from 35% to 56% after ligation in the minority of studies that included internal carotid imaging [58].

In post-discharge imaging after carotid preservation or reconstruction, mostly obtained using available imaging rather than scheduled surveillance, incidental findings of mild/moderate stenosis, intimal irregularities, or small pseudoaneurysms are common [4,19,56]. The increasing frequency of anomalous carotid imaging findings over time may represent changes in patient selection, secular improvements in imaging technology, and/or wider variation in the type and timing of post-reconstruction imaging; regardless, taken together these data demonstrate that reconstruction does not guarantee a patent or normal carotid artery. It is worth noting additionally that the clinical significance of mild or moderate post-reconstruction stenosis, intimal irregularities, and small pseudoaneurysms are unclear, and there is no standard management of these imaging findings. Studies that describe these abnormalities either explicitly report that they were not treated or provide no information on whether treatment was pursued [19,57].

Initial investigations of carotid reconstruction in the early 1990s highlighted the potential for reduction in the prevalence of right-hemispheric neuroimaging abnormalities in patients with reconstructed versus ligated vessels, but the findings of more than a dozen investigations of head/brain CT, MRI, and/or EEG studies over the last 3 decades are highly heterogeneous [4,56]. In sum, the existing literature on does not show any consistent association between carotid preservation and burden or sidedness of subclinical stroke, perfusion asymmetry, brain volume, or subclinical seizure.

Existing literature also fails to show a convincing difference in either short- or long-term neurologic or neurodevelopmental

outcomes between ligation and preservation or reconstruction [19,56,58]. Long-term neurologic sequelae of neonatal and pediatric ECMO cannulation are common, but it is difficult to separate the neurologic effects of critical illness and ECMO pump dynamics from cannulation- or reconstruction-specific factors without prospective comparison of cannulation techniques.

Repeat cannulation following cervical vessel reconstruction is a more concrete and uniformly defined outcome. Among patients who require second ECMO runs following initial decannulation, rates of successful repeat cervical ECMO cannulation after vascular repair or reconstruction exceed 85%, though most available data come from small retrospective case series and no studies directly compare vessel sparing, reconstruction, and ligation [19]. Young age, shorter time on ECMO, and anticipated need for future ECMO – particularly relevant in children with congenital cardiac anomalies, of whom 5% will require repeat ECMO cannulation in the future, versus 1%–2% of children without congenital cardiac disease – are all proposed considerations for preserving or reconstructing rather than ligating cervical vessels, though none of these factors have been systematically or prospectively investigated as predictors of benefit from differential vessel management.

4.4. Mid- and long-term cerebrovascular imaging and surveillance

Only half of ECMO practitioners who frequently preserve cervical vessels obtain routine vascular imaging after hospital discharge [4], and current data on the long-term vessel imaging features is exclusively comprised of single-center studies that use available imaging findings obtained outside the context of ECMO surveillance. Despite a high prevalence of vessel anomalies in these studies, the need for subsequent imaging to surveil incidentally discovered vessel abnormalities is not clear [19]. Increased left carotid intimal thickness among adolescents who had undergone cervical ECMO cannulation and right carotid ligation was reported more than a decade ago, and is hypothesized to be a consequence of persistently increased compensatory contralateral flow [21]. In adults with subclinical cardiovascular disease, carotid intimal thickness is a marker for risk of stroke and clinically significant carotid artery stenosis, but whether this finding in adolescent ECMO survivors is associated with similar long-term risks remains to be determined. The role of contralateral duplex or intracranial vascular imaging have not been elucidated regardless of carotid ligation versus reconstruction.

No specific society guidelines exist for vessel surveillance among pediatric ECMO survivors, regardless of cannulation site, ligation, or reconstruction. ELSO guidelines updated in 2021 recommend in-hospital baseline and follow up neurologic assessments after neonatal and pediatric ECMO decannulation [18]. Although these guidelines provide a detailed schedule for neurologic surveillance after ECMO, they not include recommendations for post-discharge carotid imaging [18]. There is no consensus on a protocol for cerebrovascular imaging after cervical ECMO, and variation exists even within centers that frequently perform vessel sparing or reconstruction. Most institutions describe duplex imaging before discharge and at least once after discharge, but timing and subsequent frequency are generally not reported in existing literature [4].

5. Femoral ECMO cannulation, decannulation, and vessel management

5.1. Cannulation

As with cervical cannulation, femoral VV or VA ECMO cannulation can be performed via open, percutaneous, or hybrid approach. Ultrasound-guided percutaneous cannulation is

common in adults and older children, and single-center studies of percutaneous cannulation in children and adolescents have demonstrated high success rates and complication rates comparable to open cannulation [43]. Multiple groups propose a hybrid access strategy using a cutdown and modified Seldinger approach in which the cannula is first tunneled through the soft tissue before being advanced over a wire into the vessel, allowing for direct visualization and precluding the need to tunnel under ultrasound guidance [13,44]. No studies in children directly compare access-related complication rates between percutaneous, hybrid, and open femoral approaches, and failed percutaneous cannulation requiring conversion to open is often not clearly differentiated from primary open approach in registry studies or single-center case series, and thus the literature on the safety and success of this approach in children is difficult to interpret.

If possible, venous and arterial cannulation for femoral VA ECMO should be performed on contralateral limbs. Venous femoral cannulation is associated with ipsilateral lower extremity venous congestion in both adults and children [59]; as limb ischemia is increased with femoral arterial cannulation (which is mitigated with placement of a distal perfusion catheter [60]) it seems prudent to avoid any additional ischemic threat by obstructing venous return in the ipsilateral limb. In extremities with limited arterial flow, venous congestion poses a theoretical risk of worsening or accelerating ischemia or compartment syndrome. Several studies have shown the benefit of distal perfusion cannula placement to reduce the incidence of distal ischemia, regardless of open versus percutaneous approach [13,43]. Since femoral venous cannulas do not typically cross the right atrium, fluoroscopy is not routinely required, and plain film X-rays can be used to confirm cannula position before circuit initiation.

5.2. Chimney graft femoral arterial cannulation

Chimney graft cannulation technique involves end to side anastomosis of a prosthetic graft (typically Dacron) to the CFA, then direct end-to-end cannulation of the graft (Fig. 9). The ECMO cannula should be tunneled through the skin and soft tissue for stability, then depending on available equipment can be secured to the graft using dedicated connectors or ties. As in other cannulation approaches, additional interrupted sutures should be used to secure the proximal catheter to the skin. In adults, chimney grafts are typically reserved for adults with small native arteries, or those with known peripheral artery disease who have either native vessel calcification that limits ECMO access or clinically significant stenosis that prevents them from tolerating relative low-flow states. Chimney grafts negate the need for distal perfusion catheter (DPC) placement, though the frequency of extremity hyperperfusion and wound complications are higher with chimney graft technique than with direct arterial cannulation with or without a distal perfusion catheter. As with most ECMO cannulation techniques, there are limited data on the use of chimney grafts in pediatric populations.

5.3. Use of distal perfusion catheters

Distal perfusion catheters are small-bore arterial catheters placed distal to the ECMO cannula site to provide arterial flow to the cannulated extremity, typically in antegrade fashion (Fig. 10). In adults, the routine use of prophylactic DPCs during femoral VA ECMO is associated with 50% or greater reduction in reported rates of acute limb ischemia in small samples, though data on efficacy and safety of DPC are drawn almost exclusively from non-randomized single-center series [13,16,61–63]. Prophylactic DPCs are malpositioned in up to 10% of adult patients, either due to

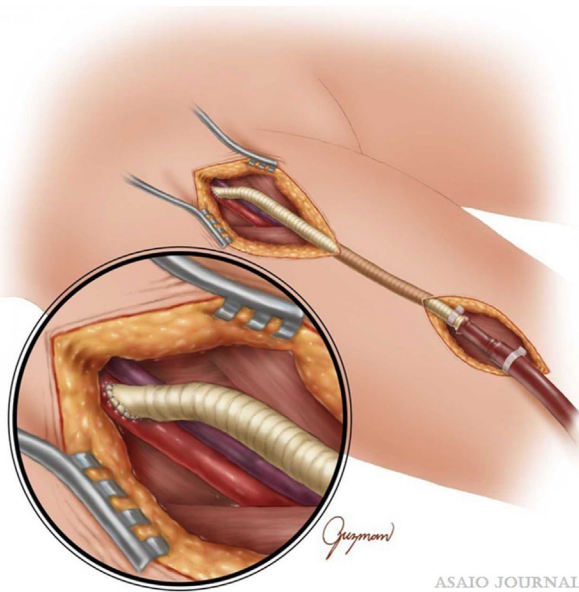


Fig. 9. Femoral Arterial Cannulation via End-to-Side Chimney Graft. A Prosthetic (Dacron or Gore-Tex) "chimney graft" is anastomosed in end-to-side fashion to the anterior surface of the common femoral artery during open exposure; the graft is then tunneled through the subcutaneous tissue to a distal site, where the graft is cannulated using an appropriately sized connector. Both incisions are temporarily closed until decannulation. Fraser et al., 2017 [13].

inadvertent intravenous placement or extravascular placement or dislodgement, and malpositioned DPCs pose a major risk of limb threat secondary to hyperperfusion and/or compartment syndrome. Indication-driven DPC placement for signs of limb ischemia are used at many centers, though come with the additional risk of reperfusion injury and resulting compartment syndrome if the diagnosis of limb ischemia is delayed [62]. In the absence of randomized, prospective data, prophylactic DPC use in adults has been implemented routinely in some centers but is not universally recognized as standard of care [64].

The benefits of prophylactic DPCs in pediatric patients are even less clear. One single-center retrospective review of 29 children undergoing femoral cannulation who received either prophylactic or reactive DPCs found fewer ischemic complications in the group undergoing prophylactic DPC placement and that reactive DPC placement only resulted in limb salvage in 50% of patients [15]. Though there was a non-statistically significant lower rate of intervention in the prophylactic DPC group (12% requiring fasciotomy versus 24% in the reactive DPC group), one patient in the prophylactic DPC group still required an above knee amputation because of ischemia sustained during their ECMO course, representing an amputation rate of 6% in this small sample.

There are no high-quality data in pediatric patients to describe the incidence of misplaced DPCs or DPC-related vascular injury but given the technical challenges of smaller vessel cannulation these rates are likely similar to or higher than those reported in adults. Utilization of appropriately placed DPCs in children is further complicated by high reported rates of DPC thrombosis or occlusion and a paucity of data on optimal DPC flows. While recent efforts to develop age-based nomograms for femoral vessel size in pediatric patients can provide some guidance for ECMO site and/or cannula selection in young children, there are not clear population-level data on lower extremity vessel size in school-aged children or for average lower extremity flows in children of any age [65]. A baseline flow of 240 cc/min has been proposed as sufficient for lower extremity perfusion in a heterogeneous pediatric population, but

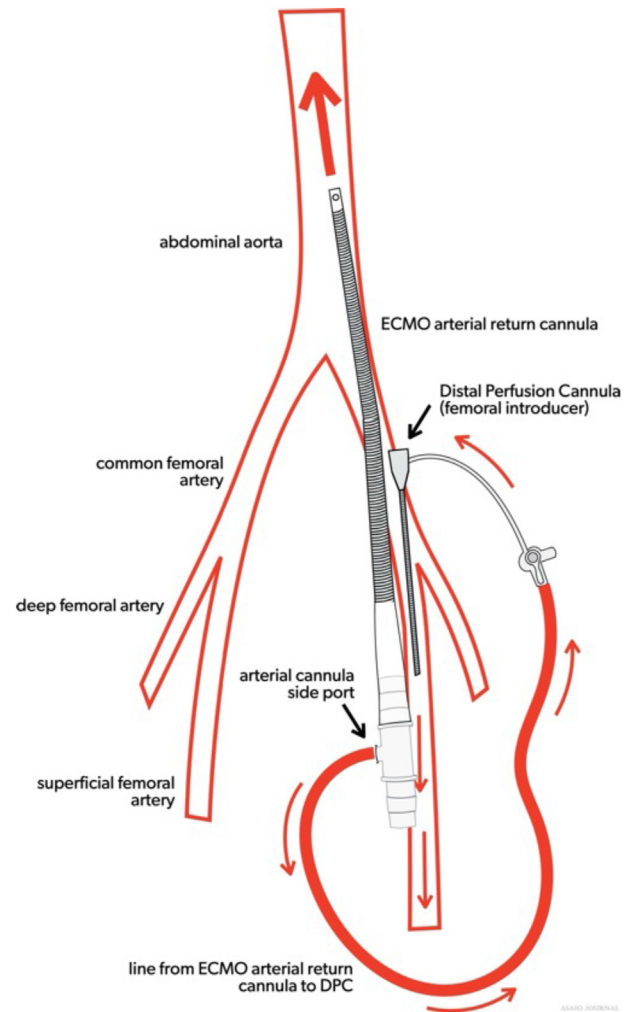


Fig. 10. Antegrade Distal Perfusion Catheter Positioning in Femoral Arterial Cannulation. Gajkowski et al. 2022 [23].

clinical and pulse exam are used to evaluate for ischemia and titrate DPC flows. Both hypo- and hyperperfusion pose risk of limb complications, and hyperperfusion may be more difficult to detect early based on clinical exam. Overall, better age-standardized reference data on normal flows in children and adolescents and on complications of DPC placement in this population are necessary to help guide DPC practice.

DPC placement complications and rates of acute limb ischemia appear similar between antegrade (via the superficial femoral artery) versus retrograde (via the dorsalis pedis or anterior tibial artery). No studies directly compare these methods in children, but most authors agree that antegrade catheters are easier to place because of vessel size, proximity to the retrograde arterial cannula access field, and higher likelihood of at least some preserved flow in proximal vessels among patients in whom indication-driven (rather than prophylactic) DPCs are used. In adults, both antegrade and retrograde DPC catheters are typically 5Fr to 7Fr in diameter; these sizes are appropriate in adolescents, but smaller catheters in younger children can provide adequate flows if necessary.

Safe DPC placement in children is best achieved using a hybrid rather than purely percutaneous technique to directly visualize vessels during cannulation and confirm that the ECMO catheter access site is appropriately proximal. Regardless of approach, accessing the superficial femoral artery with an antegrade wire

before placing the retrograde common femoral ECMO cannula allows distal vessel access to be confirmed while femoral flow is still intact [13]. By this method, common femoral artery ECMO cannula placement can proceed and, if necessary, DPC placement can be delayed until after systemic ECMO flows have been initiated. A single arteriotomy for both ECMO cannula and DPC or two separate arteriotomies, one for each catheter, have been described. Theoretically, a single arteriotomy is simpler to repair and allows for easier confirmation of intra-arterial DPC placement but may also result in more cannula site bleeding; there are no data to support one technique over the other.

Frequent examinations for signs of edema or ischemia, pulse exams, and extremity pulse oximetry are universally recommended in patients undergoing femoral VA ECMO. Most centers use nurse-driven limb monitoring protocols that include clinical exam and doppler pulse checks at least hourly while ECMO cannulas are in place, paired with serial lab evaluation to include markers of rhabdomyolysis (myoglobin, creatinine kinase) that might not otherwise be collected as part of standard ECMO monitoring [13]. Routine calf near infrared spectroscopy (NIRS) to monitor extremity perfusion has also been proposed by some high-volume adult ECMO practitioners, with a threshold of skin capillary SpO₂ below 40% or 25% or greater decline from baseline used trigger duplex evaluation and/or intervention [60].

5.4. Decannulation

Cannula placed by open technique, including chimney graft access, should be removed in the operating room with direct repair or reconstruction and balloon thrombectomy if significant distal thrombus is present. It is reasonable to perform a primary repair after debridement if the arteriotomy is small and the proximal vessel is healthy; however, the proximal intima can be difficult to assess, and intimal damage increases with time on ECMO. Primary repair is a less viable option as the duration of femoral ECMO therapy increases. There are no consensus guidelines for the specific duration beyond which angioplasty or graft reconstruction should be considered, but some experts suggest 2 weeks as an approximate threshold [66]. Similarly, no comprehensive set of indications exists to guide the use of patch or graft reconstruction versus primary closure of the arteriotomy. Given the paucity of data on potential consequences of femoral artery stenosis in young patients, even subtle narrowing with primary repair should prompt consideration of a patch angioplasty (Fig. 11). Ipsilateral greater saphenous vein or bovine pericardium have both been described as appropriate conduits.

Known local dissection or concerns about intimal injury—either based on intraoperative findings or on clinical history of long ECMO course, large cannula relative to vessel size, or difficult

cannulation—are potential indications for interposition graft repair. As in non-iatrogenic trauma, ipsilateral autologous greater saphenous vein is the preferred conduit, though better long-term data on interposition vein outcomes in pediatric populations is needed. If the greater saphenous vein is of insufficient size, the deep femoral vein is a viable alternative conduit. Cannulas placed via chimney graft technique should be removed in the operating room with suture ligation of the chimney graft near the graft-arterial anastomosis, and excess graft should be excised.

Percutaneously placed catheters may be safely removed at bedside with manual compression followed by ultrasound-confirmed placement of a mechanical compression device for 10–12 h (e.g., FemoStop II Plus; Radi Medical Systems, Uppsala, Sweden) [13,43]. In adult populations percutaneous techniques (either with pressure or with percutaneous closure devices) are associated with shorter procedure times and lower rates of cannulation site infections [67], but many groups choose to routinely decannulate via open surgical technique in children in order to evaluate and primarily repair femoral arteriotomies [60]. The use of endovascular pre-closure or closure devices is not standard in children given the limited experience with these devices among pediatric surgeons but is a target for future study as the role of non-surgeon pediatric interventionalists in ECMO cannulation expands.

Frequent site and distal neurovascular checks are necessary to ensure that bleeding or ischemia after decannulations are recognized immediately. In addition, limbs are at risk for reperfusion injury that may not manifest until hours following cannulation and may require urgent fasciotomy. Monitoring of serum creatinine kinase levels may be helpful to assess for muscle injury. Though there are no consensus guidelines to dictate the specific frequency of checks or duration of compression, pediatric ECMO practitioners can extrapolate from adult endovascular protocols. Typically, nurse-driven protocols for site and distal pulse exams every 15 min for the first 1–2 h, then gradually spaced to 30 min and 1 h over the first 6–12 h post-decannulation allow for early detection of clinically significant hemorrhage or ischemia. These should be paired with head-of-bed flat and log-rolling restrictions for 4–6 h after decannulation to prevent early re-bleeding or thrombus.

5.5. Common femoral cannula-related complications and their management

Limb ischemia is the most reported acute complication of femoral VA ECMO cannulation. Most limb ischemia is the result of thromboembolic events, though arterial dissection, pseudoaneurysm, and rupture of the common femoral artery (or, less commonly, the ipsilateral external or common iliac arteries) have also been reported in 1%–4% of adult femoral ECMO cannulations, with no clear associations between open versus percutaneous cannulation

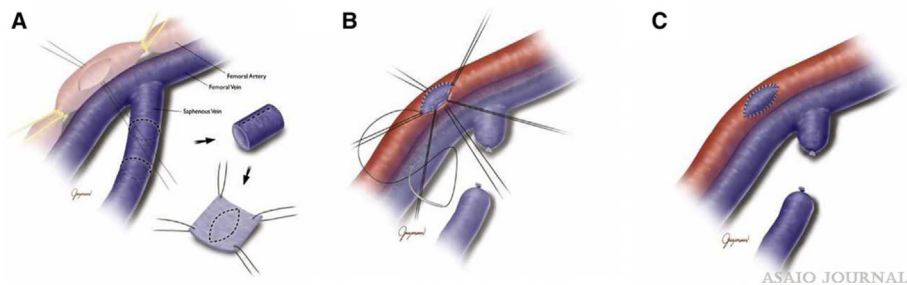


Fig. 11. Common Femoral Artery Patch Repair at time of Decannulation. (A) Ipsilateral saphenous vein is harvested for patch creation. The femoral artery is dissected over a length sufficient to achieve proximal and distal control, and the existing arteriotomy is debrided to healthy edges, extending longitudinally if necessary. (B) The vein patch is trimmed to size and anastomosed to the arteriotomy in a running fashion. (C) Completed femoral artery patch repair. Fraser et al., 2017 [13].

approach and ischemic or other vascular complications [62,64,68]. Despite high rates of ischemia, fasciotomy and lower extremity amputation are relatively uncommon complications in adults—1%–10% and 0%–2.3%, respectively [68]. Though ELSO registry data report rates of lower extremity ischemia, fasciotomy, or amputation are close to 0% among children undergoing femoral VA ECMO, single-center series in pediatric populations suggest the incidence is closer to 5%–7%.

Dissection and pseudoaneurysm during or after ECMO decannulation in children are rare but can be managed similarly to those in adults and should be co-managed with the help of vascular surgery. For pseudoaneurysms, ultrasound-guided compression and/or thrombin injection is appropriate for pseudoaneurysms that are small, have a narrow neck, and are not causing distal ischemia. Size greater than 2 cm (or twice the diameter of the native femoral artery) is an indication for operative repair.

Late complications of lower extremity ECMO cannulation are not well studied in children. Subclinical stenosis and reduced arterial flow are common after VA ECMO, but clinically significant or progressive ischemia appears rare. One single-center case–control series of children with a history of diagnostic femoral arterial cannulation found reduced mean arterial brachial index and 37% prevalence of femoral stenosis or occlusion at 5–14 years post-procedure, but only 2% of patients had symptomatic claudication or limb length discrepancy [69]. Lack of standardized follow up of patients (and limbs) after ECMO decannulation limits even observational studies, and the relative paucity of data on other non-ECMO pediatric vascular outcomes prevent comparisons to similar populations.

5.6. Mid- and long-term lower extremity vascular imaging and surveillance

As with cervical ECMO cannulation, there are no consensus guidelines for clinical and imaging surveillance following pediatric femoral ECMO decannulation, and most reported protocols are extrapolated from pediatric vascular trauma, adult vascular surgery, and/or adult ECMO literature. For patients who undergo decannulation with primary or patch arterial repair, many centers perform arterial duplex before discharge and repeat imaging and clinical exam every 3 months for the first year after decannulation, with additional imaging as needed to follow abnormal findings [13].

6. Alternative cannulation sites

6.1. Subclavian or axillary artery cannulation

Axillary artery cannulation for VA ECMO via direct arterial cannulation or right axillary artery chimney graft has been described in adults since the early 2000s and is used in up to 30% of adult VA ECMO runs at some centers, generally for patients with difficult vascular access or contraindications to femoral cannulation [70,71]. Open exposure of the right axillary artery is performed at the deltopectoral groove. In direct cannulation, the cannula can be placed via Seldinger technique with or without a pre-placed purse-string, or via transverse arteriotomy. The cannula tip is advanced just proximal to the takeoff of the right common carotid artery before the cannula is secured, either using an existing purse-string or using circumferential arterial ties and a bumper. A 3Fr to 5Fr diameter distal perfusion catheter is typically inserted to restore proximal right upper extremity flow. To improve cannula and DPC stability, both can be tunneled through soft tissue before insertion. An alternative chimney graft technique is also described, in which a synthetic Dacron graft is sewn in an end-to-side fashion to the

axillary artery through the same open approach, and the graft itself is tunneled through the soft tissue to a separate, adjacent incision site. The ECMO cannula is advanced through the graft until the cannula tip sits just proximal to the graft-arterial anastomosis and secured to the graft using either silk ties or a dedicated connector. In either technique, the primary exposure incision is then temporarily closed, and the proximal cannula is secured to the skin with several interrupted sutures. While subclavian cannulation through an infraclavicular exposure has also been described, axillary access is generally preferred because it allows for the insertion of a longer stretch of cannula, providing better stability with lower risk of covering the origin of the right common carotid [72].

A higher incidence of extremity hyperperfusion in axillary versus femoral arterial cannulation has been reported, though studies directly comparing the two strategies are lacking. Upper extremity amputations complicating axillary cannulation are exceedingly rare based on existing case series, and axillary and femoral cannulation in adults appear to have comparable overall ECMO survival outcomes. Axillary cannulation has not been routinely performed in children, and thus pediatric-specific complication rates for axillary arterial ECMO cannulation are not available.

6.2. Extraperitoneal iliac or inferior vena cava

Case reports of direct iliac and/or IVC cannulation in neonates and children are rare and mostly limited to infants with aortic or central venous anomalies that prevent traditional cervical cannulation [41,73]. A paramedian incision over the right anterior abdominal wall between the level of the umbilicus and the anterior superior iliac spine can be used to access the retroperitoneal space through an extraperitoneal plane. Depending on vessel size and flow needs, this dissection is then continued until the external iliac vessels and/or IVC are identified. A small oblong purse string suture is placed at the planned arterial or venous cannulation site using 5–0 or 6–0 Prolene. The cannula is then inserted through the purse string and positioned under fluoroscopy before it is secured with a Rummel tourniquet [41]. Arterial reconstruction with patch angioplasty is performed at the time of decannulation; some narrowing of the IVC and/or iliac vessels is generally tolerated, and venous decannulation can be achieved without reconstruction by cinching the pre-placed purse-string at the time of cannula removal [41,73].

6.3. Cannulation for extracorporeal cardiopulmonary resuscitation

Extracorporeal cardiopulmonary resuscitation (ECPR) describes cannulation for and initiation of ECMO in the setting of active cardiopulmonary arrest. Though general principles for ECMO cannulation still apply in the context of cardiopulmonary resuscitation, there are additional considerations specific to ECPR. In neonates, open cervical approach remains the mainstay because vessel size and flow limitations preclude femoral access, and because the vast majority of ECPR requires VA support. In older, larger children who are potential candidates for either cervical or femoral cannulation, active CPR may interfere with the cervical field and make femoral cannulation preferable. Percutaneous cannulation, even with US guidance, can be more challenging in pulseless and/or hypovolemic patients, and open or hybrid approaches may be faster. These stipulations are hypothetical, and no published studies directly compare time to cannulation, success rates, and cannulation-related complications between site and/or approach.

Few studies have examined cannulation strategies specifically in the context of ECPR. One large single center retrospective review of ECPR in children with congenital cardiac anomalies found no

differences in ECMO survival between central, cervical, and femoral cannulation sites, though it did not examine the role of open versus percutaneous strategies [74]. Another single-center series of percutaneous femoral cannulation in older, larger children (most >7 years of age and >20 kg) found to 100% success rates within a subgroup of patients undergoing ECPR performed by pediatric interventional cardiologists, but no open or hybrid comparison group was included [43]. Rapid response and a protocolized team approach to ECPR is the most crucial factor for successful cannulation, and the best cannulation strategy is the one most preferred by the physician(s) performing cannulation.

7. Additional considerations for ECMO cannulation and vessel management

7.1. Who should cannulate?

General pediatric surgeons and congenital cardiovascular surgeons perform the vast majority of ECMO cannulation in children. By contrast, medical intensivists and interventional cardiologists perform a large share of peripheral ECMO cannulation in adult populations, and do so with high success and low complication rates that are comparable to those of surgical specialties [75]. Based on data supporting the feasibility of safe and timely cannulation by non-surgeons, ELSO consensus statements encourage qualified intensivists to perform percutaneous cannulation in adults as long as surgical support for emergent vascular or intrathoracic complications is available [75]. More routine use of percutaneous access in adults than in children, lower overall access-related complication rates in adults, and potentially broader experience with ultrasound guided vascular access among adult compared to pediatric medical intensivists limit the generalizability of these protocols to pediatric populations.

No studies directly compare peripheral cannulation outcomes by physician specialty in children. One series of percutaneous emergent VA ECMO access by interventional cardiologists reported 100% success and a single incident major vascular complication (a femoral artery avulsion injury that was managed endovascularly at the time of cannulation and then repaired surgically at decannulation) in 23 consecutive femoral cannulations, with overall outcomes similar to historical institutional controls [43]. Surgeons participated in developing the protocol reported in this case series, and open decannulation plus/minus arterial repair or reconstruction by surgeons was performed in half of the electively decannulated patients in the cohort. Regardless of physician specialty, existing guidelines emphasize a multidisciplinary approach to decisions about ECMO timing, cannulation site and cannula size, and modality. Anecdotally, many centers use unit-based protocols (e.g., congenital cardiovascular surgeons are the first line consult for ECMO cannulation in children in pediatric cardiac intensive care units, whereas general pediatric surgeons are consulted by default for ECMO in non-cardiac intensive care units), though not all pediatric ECMO centers are fully staffed by pediatric cardiovascular surgeons. Protocols for decannulation are often less explicit or specific than those for cannulation, and this may contribute to variation in staffing and decannulation technique [6,43].

Involvement of vascular surgeons in pediatric ECMO cannulation or decannulation is a specific point of variation. Vascular surgeons are not universally available for urgent emergent procedures at all children's centers, especially at stand-alone children's hospitals. Understanding whom to reach and how to reach them may be difficult if a protocol for pediatric vascular consults is not in place, even at centers with on-site or nearby adult hospitals with vascular surgery coverage. A recent survey of the American Pediatric Surgery Association (APSA) found pediatric surgeons in

practice for fewer than 10 years were more than twice as likely to consult vascular surgery than their more senior colleagues [6]. This suggests pediatric ECMO will be an increasingly relevant subject for the practicing vascular surgeon, and that divisions of pediatric surgery will benefit from collaborations with adult vascular surgery to promote optimal outcomes.

7.2. Where should cannulation take place?

Pediatric peripheral ECMO cannulation is performed in the operating room in neonatal or pediatric intensive care units. Cannulation setting varies by institutional resources and protocols, by ECMO type, and patient acuity. Bedside cannulation in the intensive care unit often precludes the need for transport of an unstable patient but may also come with space and equipment constraints. With the exception of bicaval VV ECMO cannulation, which appears to be safer when performed in the operating room or hybrid suite under fluoroscopy [34,35], there are no clear data to suggest that site of cannulation is independently associated with access complications [14].

Emergency department cannulation for unstable patients—typically those presenting with refractory out of hospital cardiac arrest or severely injured trauma patients—is an expanding practice in adult populations but is not well-studied in children. Similarly, protocols for out-of-hospital ECLS in adults with refractory cardiac arrest have demonstrated pre-hospital cannulation success rates approaching 100% [76,77]; however, these programs exist only at highly select centers, and patients <18 years of age have been excluded from all major trials of this approach. Several high-volume pediatric ECMO centers have mobile ECMO units, as part of which an ECMO team will travel to hospitals without ECMO capabilities and perform cannulation on-site for patients otherwise too unstable for transport. These patients are then transported on ECMO support of a mobile circuit to the team's home institution [78]. Though relatively uncommon, this practice results in similar survival outcomes when practiced by experienced teams [79].

7.3. How should cannula position be confirmed and surveilled?

Regardless of modality and approach, peripheral ECMO cannula position should be radiographically confirmed by portable X-ray or fluoroscopy, depending on setting. The tip of a cervical arterial cannula should sit in the right common carotid just distal to its takeoff from the innominate artery, between the first and second ribs on portable anterior-posterior chest X-ray. A femoral arterial cannula tip should terminate in the abdominal aorta or distal external iliac artery. Ideal venous drainage cannula positioning in multi-site VA or VV ECMO depends on catheter type—single drainage port catheters should terminate in the low right atrium or near the inferior cavoatrial junction, and catheters with multiple drainage ports (typically delineated by radiopaque stripes across the length of the drainage port segment) should span the cavoatrial junction [80].

As noted previously, both echocardiographic and fluoroscopic image guidance are strongly recommended during placement of bicaval dual-lumen cannulas to reduce the risk of major complications [34–37]. Variation in radiographic markings among dual-lumen venovenous cannulas is important to note: bicaval cannulas tend to be radiopaque at the tip of the reinfusion port(s), which should be positioned in the right atrium with port markers oriented toward the tricuspid valve. The dual-lumen VV cannula drainage lumen extends 3–5 cm beyond the reinfusion port tip and should be positioned in the hepatic IVC. This lumen is often radiolucent, and some but not all dual-lumen cannulas include a radiopaque BB marker at the distalmost tip of the drainage lumen.

Knowledge of institutional cannula characteristics, placement under fluoroscopy and echocardiography, clear documentation of the location of the distal drainage tip of dual-lumen cannulas at the time of cannulation, and close communication with radiology colleagues are necessary to identify early or subtle cannula migration and minimize unnecessary cannula revisions. Cannula repositioning, especially bicaval VV cannulas, is a risk factor for vascular injury [33,35], and if repositioning a wire-reinforced dual-lumen cannula is necessary it should be performed under similar dual modality imaging guidance as is used during initial cannulation [34].

A chest X-ray should be obtained immediately after cannulas are secured and dressings are applied to confirm stable positioning and establish a baseline for serial imaging during ECMO support. At minimum, a daily chest radiograph should be obtained to ensure stable cannula positioning; this also serves to screen for other indicators of iatrogenic injuries such as hemothorax or pneumothorax [81]. While some centers routinely perform a daily echocardiogram to confirm dual-lumen cannula positioning, there is no evidence-based interval for echocardiographic surveillance. Any clinical signs concerning for cannula migration (e.g., change in external cannula position, acute change in flows, clinical deterioration, or new evidence of cerebral or extremity hypo- or hyperperfusion) should prompt a STAT repeat radiograph. Duplex ultrasound—either formal or bedside—is a useful adjunct to evaluate intravascular positioning of axillary or femoral cannulas and can detect associated occlusions or thrombus, though routine use of ultrasound for surveillance in the absence of clinical concerns for migration is not standard [13,59].

7.4. Post-ECMO anticoagulation

Current ELSO anticoagulation guidelines do not address the post-ECMO period [82]. Routine anticoagulation is not recommended after cervical vessel ligation or percutaneous femoral decannulation without reconstruction. In cases of open vessel-sparing cannulation or vascular reconstruction, regardless of site, reported practices for anticoagulation vary widely; even at centers that routinely repair the carotid and/or jugular vessels after cervical cannulation, fewer than 1/3 have a standard anticoagulation protocol [4]. The most commonly described anticoagulation regimens following both carotid and femoral reconstruction include short-course intravenous heparin therapy [55] with or without extended subcutaneous heparin or aspirin, generally one to three months unless there is clinical or duplex concern for thrombus [13,53,54].

8. Conclusions

Existing literature on best-practices in vessel management during and following pediatric ECMO is limited by a lack of prospective or randomized studies and an overall paucity of long-term follow up data. Recent guidelines regarding long-term neurodevelopmental follow up may yield additional surveillance data on the prevalence of neurologic and cerebrovascular deficits. However, larger cohorts with serial bilateral carotid duplex imaging are necessary to evaluate the long-term risk/benefits of carotid reconstruction or preservation versus ligation, establish the frequency of long-term cerebrovascular pathology among the ECMO survivor population, and to provide a basis for cerebrovascular screening recommendations.

Knowledge gaps persist regarding age, weight, body mass index, vessel size, or other clinical predictors that can be used to guide best practices in site selection and cannulation strategy in older children and adolescents. More complete ascertainment of acute vascular complications in ELSO registry data may help to fill some

of these gaps, but prospective multi-institutional studies of optimal cannulation technique in older children and adolescents are sorely needed. Understanding which patients and clinical scenarios might most benefit from involvement of additional surgical subspecialties, especially vascular or cardiovascular surgery, has the potential to streamline institutional protocols and prevent clinically significant vascular ECMO complications, reduce variation in care and improve outcomes.

References

- [1] Erdil T, Lemme F, Konetzka A, et al. Extracorporeal membrane oxygenation support in pediatrics. *Ann Cardiothorac Surg* 2019;8:109–15. <https://doi.org/10.21037/acs.2018.09.08>.
- [2] Esangbedo ID, Brunetti MA, Campbell FM, et al. Pediatric extracorporeal cardiopulmonary resuscitation: a systematic review. *Pediatr Crit Care Med* 2020;21:e934–43. <https://doi.org/10.1097/PCC.0000000000002373>.
- [3] Johnson K, Jarboe MD, Mychaliska GB, et al. Is there a best approach for extracorporeal life support cannulation: a review of the extracorporeal life support organization. *J Pediatr Surg* 2018;53:1301–4. <https://doi.org/10.1016/j.jpedsurg.2018.01.015>.
- [4] Linden AF, Howk AA, Billimoria Z, et al. Neonatal carotid artery and internal jugular vein management practices at extracorporeal membrane oxygenation decannulation: No standard approach. *J Pediatr Surg* 2023;58:2196–200. <https://doi.org/10.1016/j.jpedsurg.2023.07.007>.
- [5] Drucker NA, Wang SK, Markel TA, et al. Practice patterns in imaging guidance for ECMO cannulation: a survey of the American Pediatric Surgical Association. *J Pediatr Surg* 2020;55:1457–62. <https://doi.org/10.1016/j.jpedsurg.2019.11.010>.
- [6] Garcia AV, Jeyaraju M, Ladd MR, et al. Survey of the American pediatric surgical association on cannulation practices in pediatric ECMO. *J Pediatr Surg* 2018;53:1843–8. <https://doi.org/10.1016/j.jpedsurg.2017.11.046>.
- [7] Bartlett RH, Gazzaniga AB, Fong SW, et al. Prolonged extracorporeal cardiopulmonary support in man. *J Thorac Cardiovasc Surg* 1974;68:918–32.
- [8] Bartlett RH. Extracorporeal life support: history and new directions. *Am Soc Artif Intern Organs J* 2005;51:487–9. <https://doi.org/10.1097/01.mat.0000179141.08834.cb>.
- [9] Bhatt P, Lekshminarayanan A, Donda K, et al. National trends in neonatal extracorporeal membrane oxygenation in the United States. *J Perinatol* 2018;38:1106–13. <https://doi.org/10.1038/s41372-018-0129-4>.
- [10] Sanaia Y, Khoubian JJ, Williamson CG, et al. Trends in mortality and costs of pediatric extracorporeal life support. *Pediatrics* 2020;146:e20193564. <https://doi.org/10.1542/peds.2019-3564>.
- [11] Extracorporeal life support registry international trend report. Ann Arbor, MI: Extracorporeal Life Support Organization; 2023.
- [12] ECLS registry international complication trend report. Ann Arbor, MI: Extracorporeal Life Support Organization; 2023.
- [13] Fraser CD, Kovler ML, Guzman W, et al. Pediatric femoral arterial cannulations in extracorporeal membrane oxygenation: a review and strategies for optimization. *Am Soc Artif Intern Organs J* 2019;65:636–41. <https://doi.org/10.1097/MAT.0000000000000884>.
- [14] Blakeslee-Carter J, Shao C, LaGrone R, et al. Vascular complications based on mode of extracorporeal membrane oxygenation. *J Vasc Surg* 2022;75:2037–2046.e2. <https://doi.org/10.1016/j.jvs.2022.01.078>.
- [15] Schad CA, Fallon BP, Monteagudo J, et al. Routine use of distal arterial perfusion in pediatric femoral venoarterial extracorporeal membrane oxygenation: ECMO for respiratory indications and circulatory support. *Artif Organs* 2017;41:11–6. <https://doi.org/10.1111/aor.12861>.
- [16] Tanaka D, Hirose H, Cavarocchi N, et al. The impact of vascular complications on survival of patients on venoarterial extracorporeal membrane oxygenation. *Ann Thorac Surg* 2016;101:1729–34. <https://doi.org/10.1016/j.athoracsurg.2015.10.095>.
- [17] Duggan EM, Maitre N, Zhai A, et al. Neonatal carotid repair at ECMO decannulation: patency rates and early neurologic outcomes. *J Pediatr Surg* 2015;50:64–8. <https://doi.org/10.1016/j.jpedsurg.2014.10.029>.
- [18] Ijsselstijn H, Schiller RM, Holder C, et al. Extracorporeal life support organization (ELSO) guidelines for follow-up after neonatal and pediatric extracorporeal membrane oxygenation. *Am Soc Artif Intern Organs J* 2021;67:955–63. <https://doi.org/10.1097/MAT.0000000000001525>.
- [19] Carpenter JL, Baker M, Sperberg K, et al. Common carotid artery imaging after vessel sparing decannulation from Extracorporeal Membrane Oxygenation (ECMO) support. *J Pediatr Surg* 2021;56:2305–10. <https://doi.org/10.1016/j.jpedsurg.2021.01.048>.
- [20] Kurkluoglu M, Badia S, Peer SM, et al. Patency of common carotid artery and internal jugular vein after a simple vessel sparing cannulation for extracorporeal membrane oxygenation support. *J Pediatr Surg* 2017;52:1806–9. <https://doi.org/10.1016/j.jpedsurg.2017.08.001>.
- [21] Friedlich P, Gonzalez J, Nield T, et al. Increased carotid artery intima-media thickness following venoarterial extracorporeal membrane oxygenation in the neonatal period. *J Perinatol* 2008;28:811–4. <https://doi.org/10.1038/jp.2008.107>.

- [22] Zimpfer D, Heinisch B, Czerny M, et al. Late vascular complications after extracorporeal membrane oxygenation support. *Ann Thorac Surg* 2006;81: 892–5. <https://doi.org/10.1016/j.athoracsurg.2005.09.066>.
- [23] Gajkowski EF, Herrera G, Hattori L, et al. ELSO guidelines for adult and pediatric extracorporeal membrane oxygenation circuits. *Am Soc Artif Intern Organs J* 2022;68:133–52. <https://doi.org/10.1097/MAT.0000000000001630>.
- [24] Butt WW, Chilleti R. ECMO for neonatal sepsis in 2019. *Front Pediatr* 2020;8: 50. <https://doi.org/10.3389/fped.2020.00050>.
- [25] MacLaren G, Butt W, Best D, et al. Central extracorporeal membrane oxygenation for refractory pediatric septic shock. *Pediatr Crit Care Med* 2011;12:133–6. <https://doi.org/10.1097/PCC.0b013e3181e2a4a1>.
- [26] Valencia E, Nasr VG. Updates in pediatric extracorporeal membrane oxygenation. *J Cardiothorac Vasc Anesth* 2020;34:1309–23. <https://doi.org/10.1053/j.jvca.2019.09.006>.
- [27] Cai T, Swaney E, Van Den Helm S, et al. The evolution of extracorporeal membrane oxygenation circuitry and impact on clinical outcomes in children: a systematic review. *Am Soc Artif Intern Organs J* 2023;69:247–53. <https://doi.org/10.1097/MAT.0000000000001785>.
- [28] Brodie D, Lorusso R, MacLaren G, et al., editors. *Extracorporeal life support: the ELSO Red book*. 6th ed. Ann Arbor, Michigan: Extracorporeal Life Support Organization; 2022.
- [29] Fletcher K, Chapman R, Keene S. An overview of medical ECMO for neonates. *Semin Perinatol* 2018;42:68–79. <https://doi.org/10.1053/j.semper.2017.12.002>.
- [30] Roberts J, Keene S, Heard M, et al. Successful primary use of VVDL+V ECMO with cephalic drain in neonatal respiratory failure. *J Perinatol* 2016;36: 126–31. <https://doi.org/10.1038/jp.2015.163>.
- [31] Ündar A, Wang S, Özyüksel A, et al. Pediatric devices. Mechanical circulatory and respiratory support. Elsevier; 2018. p. 271–97. <https://doi.org/10.1016/B978-0-12-810491-0.00009-6>.
- [32] Johnson SM, Itoga N, Garnett GM, et al. Increased risk of cardiovascular perforation during ECMO with a bicaval, wire-reinforced cannula. *J Pediatr Surg* 2014;49:46–50. <https://doi.org/10.1016/j.jpedsurg.2013.09.029>.
- [33] Speggorin S, Robinson S, Harvey C, et al. Experience with the Avalon® bicaval double-lumen veno-venous cannula for neonatal respiratory ECMO. *Perfusion* 2015;30:250–4. <https://doi.org/10.1177/0267659114540020>.
- [34] Berdajs D. Bicaval dual-lumen cannula for venovenous extracorporeal membrane oxygenation: Avalon® cannula in childhood disease. *Perfusion* 2015;30:182–6. <https://doi.org/10.1177/0267659114544714>.
- [35] Ramaraj A, Jensen G, Rice-Townsend S, et al. Similar frequency of atrial perforation between atrial and bicaval dual lumen veno-venous ECMO cannulas in a pediatric population. *Perfusion* 2022;37:752–6. <https://doi.org/10.1177/02676591211030767>.
- [36] Subramanian S, Vafaeezadeh M, Parrish AR, et al. Comparison of wire-reinforced and non-wire-reinforced dual-lumen catheters for venovenous ECMO in neonates and infants. *Am Soc Artif Intern Organs J* 2013;59:81–5. <https://doi.org/10.1097/MAT.0b013e31827b519c>.
- [37] Zamora IJ, Shekerdemian L, Fallon SC, et al. Outcomes comparing dual-lumen to multisite venovenous ECMO in the pediatric population: the Extracorporeal Life Support Registry experience. *J Pediatr Surg* 2014;49:1452–7. <https://doi.org/10.1016/j.jpedsurg.2014.05.027>.
- [38] Das S, Gupta S, Das D, et al. Basics of extra corporeal membrane oxygenation: a pediatric intensivist's perspective. *Perfusion* 2022;37:439–55. <https://doi.org/10.1177/02676591211005260>.
- [39] Pollack BE, Kirsch R, Chapman R, et al. Extracorporeal membrane oxygenation then and now; broadening indications and availability. *Crit Care Clin* 2023;39: 255–75. <https://doi.org/10.1016/j.ccc.2022.09.003>.
- [40] Chamogeorgakis T, Lima B, Shafii AE, et al. Outcomes of axillary artery side graft cannulation for extracorporeal membrane oxygenation. *J Thorac Cardiovasc Surg* 2013;145:1088–92. <https://doi.org/10.1016/j.jtcvs.2012.08.070>.
- [41] Harvey C. Cannulation for neonatal and pediatric extracorporeal membrane oxygenation for cardiac support. *Front Pediatr* 2018;6:17. <https://doi.org/10.3389/fped.2018.00017>.
- [42] Cairo SB, Arbutnot M, Boomer L, et al. Comparing percutaneous to open access for extracorporeal membrane oxygenation in pediatric respiratory failure. *Pediatr Crit Care Med* 2018;19:981–91. <https://doi.org/10.1097/PCC.0000000000001691>.
- [43] Buyukgoz C, Sandhu H, Shah S, et al. Strategies and techniques for percutaneous Veno-Arterial ECMO cannulation and decannulation in children. *Cathet Cardiovasc Interv* 2023;101:1088–97. <https://doi.org/10.1002/ccd.30649>.
- [44] Huang S-C, Wu E-T, Chen Y-S, et al. Extracorporeal membrane oxygenation rescue for cardiopulmonary resuscitation in pediatric patients. *Crit Care Med* 2008;36:1607–13. <https://doi.org/10.1097/CCM.0b013e318170b82b>.
- [45] Rollins MD, Hubbard A, Zabrocki L, et al. Extracorporeal membrane oxygenation cannulation trends for pediatric respiratory failure and central nervous system injury. *J Pediatr Surg* 2012;47:68–75. <https://doi.org/10.1016/j.jpedsurg.2011.10.017>.
- [46] Di Gennaro JL, Chan T, Farris RWD, et al. Increased stroke risk in children and young adults on extracorporeal life support with carotid cannulation. *Am Soc Artif Intern Organs J* 2019;65:718–24. <https://doi.org/10.1097/MAT.0000000000000912>.
- [47] O'Neil ER, Guner Y, Anders MM, et al. Pediatric highlights from the extracorporeal life support organization registry: 2017–2022. *Am Soc Artif Intern Organs J* 2024;70:8–13. <https://doi.org/10.1097/MAT.0000000000002078>.
- [48] Jensen AR, Davis C, Gray BW. Cannulation and decannulation techniques for neonatal ECMO. *Semin Fetal Neonatal Med* 2022;27:101404. <https://doi.org/10.1016/j.siny.2022.101404>.
- [49] St Louis JD, Plunkett MD. Extracorporeal cardiopulmonary resuscitation (ECPR): initiation and surgical technique in the pediatric population. *Operat Tech Thorac Cardiovasc Surg* 2019;24:176–86. <https://doi.org/10.1053/j.optechstcvs.2019.10.001>.
- [50] Bauer HK, Malone MP. Friend or foe? Venoarterial ECMO via carotid artery “jump graft”: a case series. *J Extra Corpor Technol* 2022;54:67–72. <https://doi.org/10.1182/ject-67-72>.
- [51] Dacar D, Zobel G, Kuttig-Haim M, et al. Extracorporeal membrane oxygenation without ligation of the carotid artery. *Crit Care Med* 1992;20:436–8.
- [52] Kipfmüller F, Bo B, Schmitt J, et al. Percutaneous, ultrasound-guided single- and multisite cannulation for veno-venous extracorporeal membrane oxygenation in neonates. *Pediatr Pulmonol* 2023;58:2574–82. <https://doi.org/10.1002/ppul.26555>.
- [53] Adolph V, Bonis S, Falterman K, et al. Carotid artery repair after pediatric extracorporeal membrane oxygenation. *J Pediatr Surg* 1990;25:867–70. [https://doi.org/10.1016/0022-3468\(90\)90193-D](https://doi.org/10.1016/0022-3468(90)90193-D).
- [54] Crombleholme TM. Carotid artery reconstruction following extracorporeal membrane oxygenation. *Arch Pediatr Adolesc Med* 1990;144:872. <https://doi.org/10.1001/archpedi.1990.02150320036021>.
- [55] Spector ML, Winitzer M, Walsh-Sukys MC, et al. Carotid reconstruction in the neonate following ECMO. *J Pediatr Surg* 1991;26:357–61. [https://doi.org/10.1016/0022-3468\(91\)90979-4](https://doi.org/10.1016/0022-3468(91)90979-4).
- [56] Zens T, Eldredge RS, Gill M, et al. Vascular reconstruction after cannulation for support with extracorporeal membrane oxygenation: literature review of data in the pediatric population. *Pediatr Crit Care Med* 2023. <https://doi.org/10.1097/PCC.00000000000003372>.
- [57] Levy MS, Share JC, Fauza DO, et al. Fate of the reconstructed carotid artery after extracorporeal membrane oxygenation. *J Pediatr Surg* 1995;30:1046–9. [https://doi.org/10.1016/0022-3468\(95\)90339-9](https://doi.org/10.1016/0022-3468(95)90339-9).
- [58] Gil LA, Apfeld JC, Gehred A, et al. A systematic review of clinical outcomes after carotid artery ligation versus carotid artery reconstruction following venoarterial extracorporeal membrane oxygenation in infants and children. *J Surg Res* 2023;291:423–32. <https://doi.org/10.1016/j.jss.2023.06.046>.
- [59] Simons J, Agricola S, Smets J, et al. Duplex analysis of cannulated vessels in peripheral veno-arterial extracorporeal membrane oxygenation. *Medicina* 2022;58:671. <https://doi.org/10.3390/medicina58050671>.
- [60] Lamb KM, DiMuzio PJ, Johnson A, et al. Arterial protocol including prophylactic distal perfusion catheter decreases limb ischemia complications in patients undergoing extracorporeal membrane oxygenation. *J Vasc Surg* 2017;65:1074–9. <https://doi.org/10.1016/j.jvs.2016.10.059>.
- [61] Augusto R, Silva MP, Campos J, et al. Arterial vascular complications in peripheral venoarterial extracorporeal membrane oxygenation support. *Rev Port Cir Cardio-Toracica Vasc* 2017;24:104.
- [62] Foley PJ, Morris RJ, Woo EY, et al. Limb ischemia during femoral cannulation for cardiopulmonary support. *J Vasc Surg* 2010;52:850–3. <https://doi.org/10.1016/j.jvs.2010.05.012>.
- [63] Lamb K, Cowan S, Evans N, et al. Successful management of bleeding complications in patients supported with extracorporeal membrane oxygenation with primary respiratory failure. *Perfusion* 2013;28:125–31. <https://doi.org/10.1177/0267659112464096>.
- [64] Lamb KM, Hirose H. Vascular complications in extracorporeal membrane oxygenation. *Crit Care Clin* 2017;33:813–24. <https://doi.org/10.1016/j.ccc.2017.06.004>.
- [65] Tadphale SD, Zurakowski D, Bird LE, et al. Construction of femoral vessel nomograms for planning cardiac interventional procedures in children 0–4 Years old. *Pediatr Cardiol* 2020;41:1135–44. <https://doi.org/10.1007/s00246-020-02363-6>.
- [66] Imamura M, Caldarone CA. Venoarterial extracorporeal membrane oxygenation neck cannulation: a fixation strategy without tourniquets. *Operat Tech Thorac Cardiovasc Surg* 2019;24:256–66. <https://doi.org/10.1053/j.optechstcvs.2019.09.003>.
- [67] Roberts SH, Schumer EM, Sullivan M, et al. Percutaneous decannulation reduces procedure length and rates of groin wound infection in patients on venoarterial extracorporeal membrane oxygenation. *JTCVS Open* 2024;18: 80–6. <https://doi.org/10.1016/j.jtcx.2024.01.012>.
- [68] Roussel A, Al-Attar N, Khaliel F, et al. Arterial vascular complications in peripheral extracorporeal membrane oxygenation support: a review of techniques and outcomes. *Future Cardiol* 2013;9:489–95. <https://doi.org/10.2217/fca.13.34>.
- [69] Taylor LM, Troutman R, Feliciano P, et al. Late complications after femoral artery catheterization in children less than five years of age. *J Vasc Surg* 1990;11:297–304. ; discussion 304–306.
- [70] Moazami N, Moon MR, Lawton JS, et al. Axillary artery cannulation for extracorporeal membrane oxygenator support in adults: an approach to minimize complications. *J Thorac Cardiovasc Surg* 2003;126:2097–8. [https://doi.org/10.1016/S0022-5223\(03\)00965-6](https://doi.org/10.1016/S0022-5223(03)00965-6).
- [71] Ohira S, Malekan R, Goldberg JB, et al. Axillary artery cannulation for venoarterial extracorporeal membrane oxygenation support in cardiogenic shock. *JTCVS Techniques* 2021;5:62–71. <https://doi.org/10.1016/j.jtcx.2020.10.035>.
- [72] Pisani A, Braham W, Brega C, et al. Right axillary artery cannulation for venoarterial extracorporeal membrane oxygenation: a retrospective single centre observational study. *Eur J Cardio Thorac Surg* 2021;59:601–9. <https://doi.org/10.1093/ejcts/ezaa397>.

- [73] Frenckner B, Övermo Tydén K, Frimmel V, et al. Cannulation of the external iliac artery in a newborn with interrupted aortic arch and premature closure of the ductus. *Journal of Pediatric Surgery Case Reports* 2020;59:101493. <https://doi.org/10.1016/j.epsc.2020.101493>.
- [74] Kane DA, Thiagarajan RR, Wypij D, et al. Rapid-response extracorporeal membrane oxygenation to support cardiopulmonary resuscitation in children with cardiac disease. *Circulation* 2010;122. <https://doi.org/10.1161/CIRCULATIONAHA.109.928390>.
- [75] DellaVolpe J, Barbaro RP, Cannon JW, et al. Joint society of critical care medicine-extracorporeal life support organization task force position paper on the role of the intensivist in the initiation and management of extracorporeal membrane oxygenation. *Crit Care Med* 2020;48:838–46. <https://doi.org/10.1097/CCM.0000000000004330>.
- [76] Suverein MM, Delnoij TSR, Lorusso R, et al. Early extracorporeal CPR for refractory out-of-hospital cardiac arrest. *N Engl J Med* 2023;388:299–309. <https://doi.org/10.1056/NEJMoa2204511>.
- [77] Bartos JA, Frascione RJ, Conterato M, et al. The Minnesota mobile extracorporeal cardiopulmonary resuscitation consortium for treatment of out-of-hospital refractory ventricular fibrillation: program description, performance, and outcomes. *EClinicalMedicine* 2020;29–30:100632. <https://doi.org/10.1016/j.eclinm.2020.100632>.
- [78] Ignat T, Desai A, Hoschtitzky A, et al. Cardiohelp System use in school age children and adolescents at a center with interfacility mobile extracorporeal membrane oxygenation capability. *Int J Artif Organs* 2022;45:134–9. <https://doi.org/10.1177/0391398821990659>.
- [79] Fouilloux V, Gran C, Ghez O, et al. Mobile extracorporeal membrane oxygenation for children: single-center 10 years' experience. *Perfusion* 2019;34:384–91. <https://doi.org/10.1177/0267659118824006>.
- [80] Thompson AF, Luan J, Al Aklabi MM, et al. Pediatric extracorporeal membrane oxygenation (ECMO): a guide for radiologists. *Pediatr Radiol* 2018;48:1488–502. <https://doi.org/10.1007/s00247-018-4211-z>.
- [81] Barnacle AM, Smith LC, Hiorns MP. The role of imaging during extracorporeal membrane oxygenation in pediatric respiratory failure. *Am J Roentgenol* 2006;186:58–66. <https://doi.org/10.2214/AJR.04.1672>.
- [82] McMichael ABV, Ryerson LM, Ratano D, et al. 2021 ELSO adult and pediatric anticoagulation guidelines. *Am Soc Artif Intern Organs J* 2022;68:303–10. <https://doi.org/10.1097/MAT.0000000000001652>.
- [83] Sakurai K, Singhal N. Extracorporeal membrane oxygenation in children: a brief review. *J Paediatr Child Health* 2022;58:1525–31. <https://doi.org/10.1111/jpc.16151>.
- [84] Wild KT, Hedrick HL, Rintoul NE. Reconsidering ECMO in premature neonates. *Fetal Diagn Ther* 2020;47:927–32. <https://doi.org/10.1159/000509243>.
- [85] Kilic A. Extracorporeal membrane oxygenation. *Encyclopedia of cardiovascular research and medicine*. Elsevier; 2018. p. 281–4. <https://doi.org/10.1016/B978-0-12-809657-4.99671-6>.